# From Brain Scores to Brain-Guided Distillation: Testing Brain Alignment as a Training Signal

MohammadErfan Jabbari

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**Abstract**

Brain alignment measures whether a model representation can predict recorded human brain responses to the same language stimuli on held-out examples. It is scientifically interesting because successful prediction reveals brain-relevant distinctions that are linearly accessible in the representation, while stopping short of showing that the model implements the brain's mechanism or thinks like a person. This thesis asks a stronger question: can a score that measures such correspondence also serve as a useful objective when training a smaller language model by knowledge distillation? We test the transition through five gates: controlled measurability, preservation under compression, movement under intervention, generalization across participants, and transfer to practical or independently biological targets. Trained language-model features contain measurable signal beyond specified nuisances, and compression produces a quality-entangled retention gradient, but direct brain-guided objectives do not yield a reliable target-specific improvement at valid fold or participant inference units while preserving comparable language-model quality in the tested regimes. Participant averaging changes the estimand and can expose an apparent trend that does not generalize to individual participants; stronger capacity, alternative losses, full fine-tuning, external heads, privileged gaze information, and a learnable synthetic neural proxy do not establish the missing biological or practical gain. The bounded answer is therefore that brain alignment is measurable but is not demonstrated here to be a reliable training signal. The broader methodological implication is that measurement quality, inference unit, target specificity, and downstream validation are separate requirements and must be tested separately.

## Contents

# 1 Introduction

Imagine that a person listens to a story in an fMRI scanner while a language model processes the same words. For each part of the story, the scanner provides a vector of measured responses across locations in the brain, while a chosen layer of the model provides a vector of artificial-neuron activations. An *encoding model* learns a linear map from the model activations to the brain responses using some of the story, and is then evaluated on parts of the story that were not used to fit the map. If the map predicts held-out responses, distinctions that matter for the measured brain signal are linearly accessible in the model representation. This predictive correspondence is what this thesis calls *brain alignment*.

The construction is easier to understand through one concrete example. Suppose the person hears "the child opened the present" and later hears "the mechanic repaired the engine." The model produces different activation patterns because the words, syntax, and meanings differ, while the scanner records different patterns across language-responsive locations. The encoding map is fitted across many such examples and is required to predict a brain-response pattern for a sentence or story segment it did not see during fitting. It succeeds only when the geometry of the model features contains distinctions that a simple linear readout can reuse for the recorded responses. The map is therefore a measurement instrument between two representational spaces, not a component that either the model or the brain is assumed to contain.

The phrase "predict the brain" also needs calibration. An $R^2$ score does not reconstruct thoughts, identify what a participant consciously understood, or imply that every measured voxel is language selective. It summarizes how well one predefined response target can be predicted relative to a baseline on held-out stimuli. Changing the participant pool, averaging responses before scoring, selecting a different ROI, or using a different split can change both the numerical score and the scientific population it describes. For that reason, the experimental design is part of the claim rather than a technical detail hidden behind the score.

Brain alignment is scientifically interesting because it offers a common measurement for two otherwise very different systems. Researchers can ask which model layers best predict language-selective brain responses, whether training improves the correspondence, and whether scaling or compressing a model preserves it [1–4]. The score is nevertheless narrower than its name can suggest. Successful held-out prediction does not show that a model thinks like a person, implements the same computation, or contains a generally useful cognitive representation. It shows that a specified readout extracts predictive information under a specified dataset, nuisance model, split, and inference unit. Low-level stimulus features, temporal leakage, response averaging, and language-model quality can all change the score without establishing the stronger interpretation [5–7].

This thesis tests what happens when the measurement is asked to do more than compare representations. In knowledge distillation, a smaller *student* model learns from the output distribution of a larger *teacher* model. Distillation can retain much of the teacher's predictive quality while changing the student's internal representation. For example, when the teacher assigns nonzero probability to several plausible next words, the student learns from that full

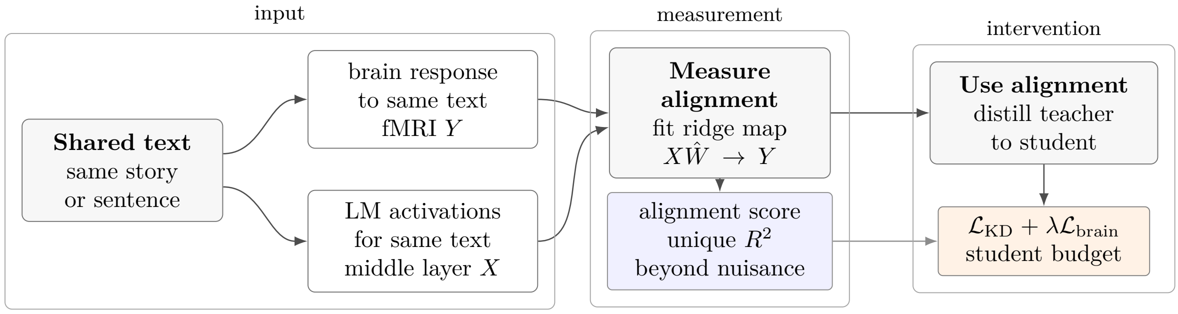


Figure 1: From measurement to intervention. A ridge encoding model maps frozen language-model activations to held-out brain responses after nuisance control. The intervention reverses the direction of use: a brain-derived target contributes an auxiliary loss while a student is distilled. The second step is scientifically stronger than the first and requires its own controls.

probability pattern rather than only from the observed next token. This transfers the teacher's output behavior, but it does not require the student's middle layers to organize information in the same way. Brain-guided distillation adds a second training signal that rewards agreement with a recorded or predicted neural target at a selected student layer. The comparison of interest is not simply whether the resulting student has a higher brain score than an arbitrary baseline. It is whether the real neural target improves held-out alignment more than a target-specific control while teacher, student, text data, training budget, and achieved language-model quality are held as comparable as the design permits. The central question is therefore whether a linear encoding score can remain scientifically and practically useful when it becomes part of the student's training objective:

> *A linear encoding model can detect brain-relevant information in an LLM representation. Does that measurement remain useful when it becomes a training objective?*

This question separates three claims that are often compressed into one. A signal can be *measurable*: a trained representation predicts held-out brain responses beyond declared nuisance features. It can be *movable*: a brain-targeted intervention changes that measurement beyond a matched control. It can be *useful*: the change survives the biologically relevant inference unit and improves an external outcome at comparable language-model quality. Measurability does not imply movability, and movability does not imply usefulness.

We organize the empirical program as five gates, shown in Figure 2. The first asks whether the signal survives controls for low-level stimulus structure, temporal leakage, and untrained-network features. The second asks how ordinary distillation and language-model quality affect the score before any brain objective is added. The third asks whether a real brain target moves the score more than a matched control target. The fourth asks whether an apparent gain survives participant-level evaluation and matched-quality comparisons. The fifth asks whether the surviving representation change improves a downstream or privileged-information task. Failure at a gate narrows what later tests can establish, but it does not erase the evidence accumulated at earlier gates.

The order matters because each gate blocks a common shortcut. A positive raw score without nuisance and untrained controls can reflect stimulus structure rather than learned language processing. A lower score after compression can reflect a worse language model rather than loss caused by distillation itself. A score that moves during training can reflect an easier target, a different perplexity, or a control loss that failed to engage. A repeatable seed effect can still disappear when participants become the units over which biological variation is measured. Finally, improvement on the same proxy used for training can fail to transfer to recorded fMRI or to a downstream language task. The thesis treats these as distinct evidential transitions instead of using one positive number to stand in for all of them.

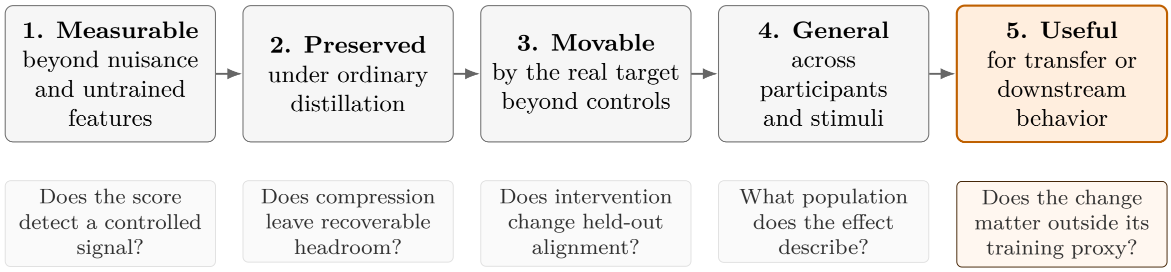


Figure 2: The cumulative experimental chain. Measurement establishes that a representation contains linearly accessible brain-predictive information; intervention asks whether a brain-targeted objective changes it; biological and practical validation ask whether the change survives the appropriate inference unit, quality controls, and an external endpoint.

The motivating hypothesis is deliberately weak. A brain objective may act as a regularizer that selects among text-compatible student representations rather than as a second teacher that supplies a new language task. This possibility is plausible because next-token objectives constrain output distributions more directly than middle-layer geometry, leaving multiple internal solutions with similar perplexity. It is not guaranteed by information theory, and the brain dataset's small and noisy supervision does not by itself establish a useful information budget. The hypothesis therefore predicts only a place to look: constrained, compressed, or under-trained regimes in which a targeted inductive bias might choose a better internal solution. The experiments, rather than the theory, must determine whether that selection occurs and whether it matters.

The work is positioned between four related literatures. Encoding-model studies measure correspondence between artificial and biological representations [1–3]. Neural-guided optimization studies use recorded or predicted responses to change learned representations [8–11]. Compression studies ask how pruning, quantization, or distillation changes alignment after training [4]. Validity studies show why nuisance subtraction, reliable splits, matched quality, and the correct biological inference unit are necessary before a score is interpreted [5–7]. The contribution is not the claim that these areas contain a single empty cell. It is a controlled test of alignment-guided distillation at their intersection, with matched controls and participant-aware evaluation used to separate measurement from intervention.

**Contributions.**

This thesis answers the cumulative questions with five contributions.

- It defines an artifact-resistant alignment estimand and verifies a trained-over-untrained signal with contiguous held-out evaluation, fold-only preprocessing, capacity-matched nuisance features, and reliability selection [E002][E006].
- It separates preservation from language-model quality, showing both alignment headroom after distillation and substantial quality entanglement across a broader model comparison [E003][E015].
- It demonstrates that the apparent intervention answer depends on the response target and inference unit: an averaged-response trend does not become a participant-level effect, and a nested sensitivity pattern does not survive randomized subset construction [E004][E005][E008][E010][E014].
- It tests capacity, loss form, voxelwise regression, and full fine-tuning, finding no reliable brain-specific improvement in the tested regimes while preserving a bounded distinction between failure of these mechanisms and a universal impossibility claim [E011][E013][E017].
- It tests downstream, privileged-information, cognitive-target, external-reproduction, and dense synthetic-target routes, identifying failed preconditions and failed transfer rather than treating an easier proxy endpoint as evidence of real-brain benefit [E009] [E016] [E019]

[E020] [E021] [E022] [E024].

The bounded empirical answer is that brain-relevant information is measurable in the tested language-model representations, but brain alignment is not demonstrated to be a reliable training signal in the tested distillation regimes. This is not a universal null about neural supervision. It is a map of where the tested causal chain holds, where it breaks, and which distinctions must remain separate: encoding quality, language-model quality, target specificity, biological inference, and downstream validation.

# 2 Related work

The literature relevant to this thesis has developed along two distinct axes: whether brain alignment is measured or optimized, and whether the model's parameter budget is preserved or reduced. Table 1 organizes these lines of work without treating adjacency as equivalence: measuring alignment after compression does not establish that alignment can guide compression, and optimizing alignment during ordinary fine-tuning does not establish that the same signal remains effective in a smaller student.

Table 1: A taxonomy of prior work by parameter-budget intervention and treatment of brain alignment. Alignment-guided distillation is the focus of this thesis.

| | Alignment measured | Alignment optimized |
|---|---|---|
| Parameter budget preserved or increased | frozen-model encoding studies [2, 12, 13] | brain-guided fine-tuning and adaptation [8, 11, 14] |
| Parameter budget reduced | post-compression measurement [4] | **alignment-guided distillation (this thesis)** |

## 2.1 Measuring brain alignment

Most brain–language-model studies treat a pretrained model as fixed and fit an encoding model from its internal representations to neural responses elicited by the same stimulus. This approach has established that alignment is structured rather than uniform across a network. Middle layers often predict language-selective activity better than early or final layers, while syntactic and semantic contributions vary across cortical regions [2, 5]. Across model families, alignment generally increases with language-model capability and scale before flattening in larger systems [12, 13]. These observations motivate the use of middle-layer representations in the present experiments, but they remain correlational: they describe where a brain score is high after training and do not show that changing the score changes the model.

Encoding results also depend on the level at which the neural target is summarized. Sentence-level ROI averages provide a compact benchmark with relatively high reliability, whereas naturalistic voxelwise datasets preserve temporal and spatial detail at the cost of harder alignment and fewer independent participants. Participant averaging emphasizes stimulus-locked structure shared across people and can improve predictability, but a high group-average score does not imply that the same relationship is strong or homogeneous for individual participants. This distinction appears throughout the broader literature because many benchmark comparisons optimize measurement reliability while intervention claims implicitly concern biological generalization. The present work treats those goals separately rather than using benchmark reliability as a substitute for participant-level inference.

The relationship between model quality and alignment is especially important for intervention studies. Better next-token prediction can accompany stronger encoding performance, so a comparison between models of unequal language quality may attribute an ordinary capability difference to a neural objective. The expanded analysis in this thesis confirms a cross-family association on the standard participant-averaged benchmark while also showing that the relationship is weaker in the capable operating range and difficult to estimate within low-signal individual participants [E015]. Accordingly, language quality is treated as a control variable rather than as evidence that alignment is either reducible to perplexity or independent of it.

## 2.2 Optimizing neural alignment

A smaller literature moves from measurement to intervention by adding neural prediction or similarity terms during model training. In speech, voxelwise objectives have been used to adapt compact encoders, with reported improvements in neural prediction and downstream speech tasks; multi-participant variants report additional data-efficiency gains [8, 9]. In text, brain-guided objectives have been applied through fine-tuning, adapters, and auxiliary prediction heads, with mixed gains across models and evaluation settings [10, 11]. Matched-quality interventions also provide evidence that deliberately increasing or decreasing alignment can change downstream behavior [14].

The positive studies differ substantially in modality, target construction, trainable component, and downstream endpoint. Some optimize a direct voxel-prediction loss, some use representational similarity, and others train a small head while leaving most of the language model fixed. A reported improvement can therefore establish that one neural objective worked under its own design without identifying a general property of "brain guidance." This thesis uses reduced external reproductions only as bounded checks of local mechanism transfer. Failure to reproduce a gain on one participant, a smaller corpus, or a different implementation is not treated as a refutation of a full multisubject result.

These studies establish that neural data can participate in an optimization objective, but they do not settle the compression question addressed here. Most preserve the base model's parameter budget and begin from a pretrained representation that is already aligned. Their positive results can therefore combine inherited alignment, fine-tuning drift, language-quality change, target learnability, and brain-specific structure. The present design separates those possibilities with cold- and warm-initialization controls, matched-quality comparisons, target-permuted twins, participant-level inference, and a downstream gate that is interpreted only after a reliable representation change is established.

Related neural-regularization studies in vision and audio illustrate why matched controls matter. Neural supervision can improve performance or robustness, but shuffled-label or non-neural objectives may reproduce part of the benefit [15, 16]. Such results do not make the neural objective uninformative; they show that a gain over an untuned baseline is insufficient to identify which part of the training signal caused it.

## 2.3 Alignment under compression

Compression studies ask a complementary question: how much alignment remains after a model is made smaller. Oota et al. [4] measure alignment after quantization and pruning and find that preservation depends on model size and compression method. This is evidence about robustness of a fixed trained representation under post-training transformation, not about using neural data to train a student.

Knowledge distillation changes the problem because it constrains a student's outputs while leaving substantial freedom in its internal representation. A student can match a teacher's next-token distribution without reproducing the teacher's middle-layer geometry, and a student trained from scratch also learns from the stimulus corpus along a path not mediated by fixed teacher features. The empirical question is therefore twofold: whether ordinary distillation preserves alignment, and whether an explicit neural term can recover or improve what is lost without merely changing language quality. The present experiments address both parts, including initialization, quality, capacity, loss-form, participant, and synthetic-target variants [E003] [E004] [E008] [E011] [E013] [E016] [E017].

This freedom creates both the opportunity and the identification problem. If several internal representations yield similar output distributions, an auxiliary target could in principle select among them without changing perplexity. If the student is under-trained or initialized differently, however, the same representational difference can arise from ordinary quality loss or inherited pretraining. Cold-start, warm-start, ordinary fine-tuning, and matched-perplexity arms are therefore not interchangeable baselines. They isolate transmission through the teacher channel, drift from a pretrained solution, change caused by more text training, and change remaining after output quality is held approximately fixed.

### 2.4 Validity, quality matching, and inference units

Brain scores are sensitive to evaluation design. Temporal autocorrelation can leak information across shuffled splits, low-level stimulus properties can support apparently positive prediction, and flexible feature spaces can exceed weaker nuisance baselines without isolating the construct of interest [6, 7]. The same scalar score may also arise from different linguistic or acoustic features across models [5]. For these reasons, held-out prediction is informative only relative to explicit splits, nuisance variables, model controls, reliability criteria, and an identified inference unit.

Three distinctions are load-bearing for the present thesis. First, contiguous sentence blocks or whole stories prevent adjacent naturalistic samples from appearing on both sides of evaluation. Second, a same-architecture untrained model and a target-permuted training twin answer different questions: the former tests what language training adds to measurement, whereas the latter tests whether an intervention depends on the correspondence between stimulus and neural target. Third, repeated folds, seeds, voxels, and participants are not interchangeable replicates. Inference over five shared stimulus folds cannot be converted into population evidence by adding seeds, spatially dependent voxels do not create independent participants, and a response averaged across people defines a different estimand from the mean effect across individuals [E004] [E006] [E008] [E014].

Quality matching completes this validity framework. Because alignment co-varies with language-model quality over the broad model range, a neural intervention must be compared with a non-neural alternative at comparable held-out quality whenever the claim is objective-specific [E015]. Matching quality still does not identify the content of an internal representation, but it removes a major alternative explanation for a higher alignment score. The contribution of the present work is therefore not the bare proposal to optimize a brain score; it is the controlled test of whether that optimization remains specific, participant-general, and useful when the model is compressed.

## 3 Methods

### 3.1 Study overview

The study follows five gates rather than treating every alignment score as the same kind of evidence. Gate 1 validates measurement: a trained language model must predict held-out brain responses beyond low-level nuisance features and a same-architecture untrained network. Gate 2 measures preservation and quality: ordinary distillation must be characterized before a brain loss is credited with changing anything. Gate 3 tests intervention: training against the real target must outperform an otherwise matched control target. Gate 4 tests biological generalization: the contrast must survive the participant, not a voxel or seed–fold cell, as the independent unit. Gate 5 tests usefulness: a reliable brain-specific representation change must improve an external endpoint at comparable language-model quality.

The experiment families instantiate these gates at increasing levels of difficulty. ROI-scale sentence experiments provide a controlled and inexpensive feasibility setting. Voxelwise naturalistic experiments test whether the measurement survives whole-story splits and subject-specific responses. Participant-level, alternative-loss, full-fine-tuning, cognitive-target, external-reproduction, and synthetic-target experiments probe distinct failure explanations. No single family is treated as a universal test of neural supervision.

### 3.2 Brain data and response targets

Functional magnetic resonance imaging (fMRI) measures changes in blood oxygenation associated with neural activity. The result at one spatial location, called a *voxel*, is a noisy time series rather than a direct measurement of individual neurons. A *region of interest* (ROI) pools voxels selected for an anatomical or functional role, such as language processing. A *participant* is one person whose responses were recorded; voxels within that participant are repeated spatial measurements, not independent biological replicates.

The blood-oxygen response is delayed and blurred relative to the words that elicited it. For isolated-sentence data, the released response already summarizes the response associated with each sentence. For continuous stories, word-level model features must be aligned to the scanner sampling grid and accompanied by delayed copies or a hemodynamic response model so that the encoding model compares compatible time points. This temporal step creates a major leakage risk because adjacent scanner samples and adjacent story segments are correlated. Holding out complete contiguous stories or blocks is therefore part of the estimand: it asks whether a mapping learned on one stretch of language predicts a genuinely unseen stretch rather than its temporal neighbors.

An ROI and a voxelwise target answer complementary questions. ROI averaging reduces dimensionality and measurement noise, which makes controlled screens feasible, but it can conceal spatially heterogeneous effects and yields only a handful of target dimensions. Voxelwise evaluation retains spatial detail and can reveal whether a pattern is widespread, but thousands of voxels from one person remain dependent measurements from that person. Neither representation is intrinsically more biologically general. Population generalization requires independent participants regardless of the spatial resolution used inside each participant.

We use two principal fMRI resources. The Tuckute benchmark contains responses to 1,000 isolated sentences in five left-hemisphere functional language ROIs [1]. Its main response is averaged across participants and voxels within each ROI, which improves signal-to-noise ratio but changes the estimand from an individual response to a group-average target. The matched mean noise ceiling for these five ROIs is a correlation-scale value of $r_{\mathrm{NC}} = 0.491$; it is reported as reliability context and is not used as a denominator for a variance-scale unique-$R^2$ score [E002][E004].

Participant averaging is performed on responses before the encoding model is evaluated. If $Y_i(S) = g(S) + \epsilon_i(S)$, where $g$ is a stimulus-locked component shared across participants and $\epsilon_i$ contains participant-specific response and measurement noise, averaging suppresses parts of $\epsilon_i$ while retaining $g$. The resulting target can be more reliable and easier to predict, but it represents the shared group response rather than the expected response of a randomly sampled individual. Because ridge fitting and $R^2$ are nonlinear operations, scoring an averaged response is not equivalent to averaging participant-specific scores. E005 and E014 use the former estimand, while E008 uses the latter.

The LeBel resource contains naturalistic story-listening fMRI from deeply sampled individuals [17]. The primary signal-validation analysis uses participant UTS03, 20 stories, and whole-story-held-out folds. Voxels are retained only when their split-half reliability exceeds 0.5 on a separate story repeated ten times, leaving 11,442 reliable voxels [E006]. The inference supported by this experiment is conditional on UTS03 and this acquisition and selection protocol. The many retained voxels increase spatial resolution but do not turn one participant into a population sample.

Reliability selection and noise-ceiling normalization are also different operations. Selection uses an independently repeated story to exclude response locations whose repeat measurements are too unstable for meaningful evaluation. Normalization would rescale a performance metric by an estimated maximum on a compatible scale. The thesis performs the former for LeBel and reports the Tuckute correlation ceiling as context, but it does not divide a variance-scale unique $R^2$ by a correlation-scale ceiling. This distinction preserves the raw experimental contrasts while preventing an apparently intuitive but dimensionally invalid "fraction of ceiling" claim.

The intervention studies use three response constructions. An averaged Tuckute target asks whether training helps predict the common response after participant averaging. Participant-specific Tuckute targets ask whether the effect generalizes across people, with the participant as the independent unit. LeBel voxelwise targets ask whether a dense within-participant regression lever takes hold under naturalistic evaluation. Because averaging occurs before the nonlinear encoding score is calculated, the averaged and participant-specific analyses estimate different quantities.

### 3.3 Alignment measurement

The concrete measurement question is whether a frozen model representation predicts brain responses for stimuli that were not used to fit the readout. Let $n$ be the number of stimulus observations, let $X \in \mathbb{R}^{n\times d}$ contain $d$ features from one language-model layer, and let $Y \in \mathbb{R}^{n\times v}$ contain responses from $v$ voxels or ROI targets. The encoding map is fitted by ridge regression:

$$\hat{W} = \arg\min_{W} \|Y - XW\|_F^2 + \alpha\|W\|_F^2. \tag{1}$$

Here $W$ maps model features to responses, $\|\cdot\|_F$ is the Frobenius norm, and $\alpha$ controls shrinkage. In plain language, the first term rewards accurate brain prediction and the second prevents a high-dimensional readout from fitting noise. The model is linear in the supplied representation, the ridge parameter is selected without using the test fold, and all learned preprocessing is fitted within the training fold. Equation 1 licenses a claim about linearly accessible predictive structure; it does not identify a shared neural mechanism.

A raw encoding score can be positive because the representation carries sentence length, timing, word identity, or other stimulus properties that also affect the brain response. The relevant question is therefore how much predictive value the contextual representation adds beyond a declared nuisance set $Z$. We estimate this increment with cross-validated semipartial $R^2$:

$$\Delta R^2 = R^2_{\text{cv}}(Y\,; [Z, X]) - R^2_{\text{cv}}(Y\,; Z)\,. \tag{2}$$

The full model receives nuisance and language-model features, while the reduced model receives the nuisance features alone. Positive $\Delta R^2$ means that $X$ improves held-out linear prediction beyond this particular $Z$. The claim is conditional on the nuisance set and evaluation design: it does not mean that every stimulus-predictable alternative has been removed.

Naturalistic stimuli create an additional shared-cause problem because even an untrained network can encode tokenization, architecture, and residual stimulus structure. We therefore define the trained-over-untrained gap:

$$\Delta_{\text{train}} = \Delta R^2_{\text{trained}} - \Delta R^2_{\text{untrained}}. \tag{3}$$

Both arms use the same architecture, nuisance features, preprocessing, and folds. The subtraction asks what language training adds over random initialization under that matched design [E006]. It removes neither every architectural confound nor every possible common cause, so the licensed conclusion remains a trained-over-untrained predictive difference rather than a mechanistic identity.

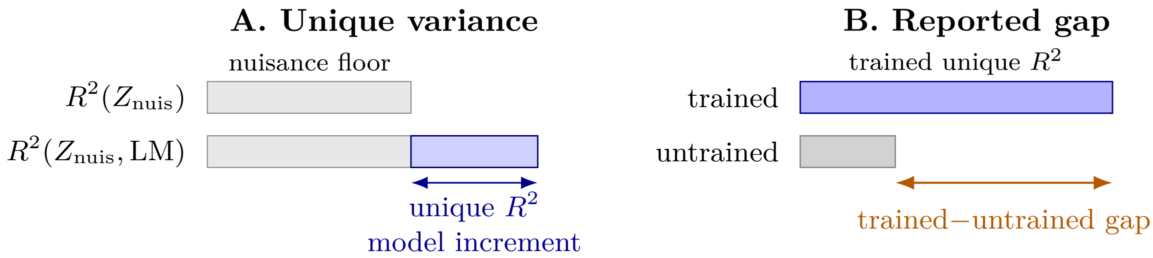


Figure 3: The two alignment contrasts. Unique variance is the held-out $R^2$ increment obtained by adding language-model features to a nuisance model. The trained-over-untrained gap subtracts the same quantity measured with a randomly initialized network of the same architecture. Bar lengths are illustrative.

Noise ceilings quantify response reliability rather than model performance. A correlation ceiling $r_{\mathrm{NC}}$ and a variance ceiling $r^2_{\mathrm{NC}}$ are different quantities and cannot be interchanged. We use reliability ceilings to define or contextualize measurable targets, but we do not divide semipartial $R^2$ by a correlation-scale ceiling.

### 3.4 Controls, uncertainty, and inference units

All time-series analyses use contiguous held-out units. The ROI benchmark uses contiguous sentence blocks, and the naturalistic analysis holds out complete stories. Randomly shuffling neighboring fMRI time points across train and test would allow temporally autocorrelated observations to leak across the split, so such splits are not used for scientific claims [6].

Nuisance capacity is matched where wide contextual and static feature blocks compete. Principal components are learned on each training fold and reduce those blocks to the same rank, while low-dimensional scalar nuisances remain unprojected. This prevents the contextual block from winning solely because it provides more columns [E006]. Surprisal is not included as a nuisance because it is itself derived from a language model and would remove part of the construct being tested; static lexical frequency remains a nuisance.

Reliability selection is independent of the evaluated stories. The voxelwise analysis uses a separately repeated story to select reliable voxels before cross-validation [17, 18]. This design avoids choosing voxels because they happened to correlate with the evaluated model. It does not establish generalization to other participants or scanners.

Intervention experiments compare a real target with a control target under matched training settings. The block-permuted twin reorders contiguous response blocks while preserving the loss shape and most optimization conditions. It is a specificity control, not a perfect randomization null, because some implemented permutations can leave observations paired with their original targets [E016]. Conclusions therefore use the observed real-minus-control contrast and do not describe the twin as exactly marginal-preserving.

The estimand, estimator, and independent unit are stated separately. For participant-specific intervention, the estimand is the expected participant-level real-minus-control change in held-out $\Delta R^2$. The estimator first averages paired seed and fold contrasts within each participant and then compares participants. The nine participants are the biological inference units; the 15 seed–fold cells within one target are repeated computational evaluations and cannot be counted as 15 independent people [E005][E008].

Confidence intervals and tests follow the independent unit available to each experiment. Participant-level tests use participants, multi-seed engineering tests use seeds, and story-level signal validation uses held-out story folds while explicitly retaining the shared-training-data dependence caveat. Power and minimum detectable effect calculations are interpreted on the same scale and unit as the corresponding test. They describe what the design could detect;

they do not convert a nonsignificant estimate into proof of zero effect.

Three examples make the distinction operational. In E004, three training seeds are evaluated on each of five held-out sentence folds; the five folds determine uncertainty about stimulus-block sensitivity because seeds within a fold share its test data. In E008, seed and fold contrasts are summarized within each person before the nine participants determine biological uncertainty. In E016, six training seeds are evaluated on the same participant-averaged Tuckute endpoint, so the sign test measures repeatability over optimizer randomness for that endpoint and says nothing about variation across people. Reporting the largest available count in each case would answer a different and usually easier question than the scientific claim.

### 3.5 Knowledge distillation and intervention objectives

Knowledge distillation asks a student to match the teacher's softened next-token distribution. Let $p_T^\tau$ and $q_S^\tau$ denote teacher and student token distributions at temperature $\tau$. The standard distillation loss is

$$\mathcal{L}_{\mathrm{KD}} = \tau^2 D_{\mathrm{KL}}(p_T^\tau \,\|\, q_S^\tau)\,. \tag{4}$$

The KL divergence penalizes token distributions that differ from the teacher, and $\tau$ exposes relative probabilities beyond the most likely token. Equation 4 constrains the student's output distribution. It does not uniquely determine the student's middle-layer geometry because many internal representations can produce similar logits.

Perplexity is the exponential of held-out mean token cross-entropy. On a shared evaluation corpus, lower perplexity means a better next-token distribution under this metric. Matched compute compares arms at the same nominal training budget, whereas matched perplexity compares them at similar achieved language-model quality. Both are reported because equal steps need not produce equal quality, and an alignment difference caused by under-training is not an objective-specific brain effect.

The primary intervention adds a brain-target regression term:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{KD}} + \lambda_B \mathcal{L}_B. \tag{5}$$

Here $\mathcal{L}_B$ is either mean-squared error to a recorded or predicted brain target or a declared contrastive alternative, and $\lambda_B$ is its weight. For a mean-squared-error (MSE) target, a learned linear prediction head maps the selected student hidden state into the target space, such as five ROI values or a dense cortical-response vector. The prediction head and the permitted student parameters are optimized jointly, but only the student is retained as the language model after training; the head is used to impose the auxiliary geometry and to check target fit. The block-permuted twin uses the same head architecture, trainable parameters, and optimizer while changing which target is paired with each stimulus. The equation asks whether the auxiliary target selects a student representation that ordinary distillation would not select. It assumes only that the two losses can be optimized jointly. It does not imply that $\mathcal{L}_B$ measures cognition, that increasing $\lambda_B$ increases alignment, or that an easier target transfers to real brain responses.

Cold-start students are initialized randomly, so any retained alignment must be learned during training. Warm-start students begin from pretrained weights and measure drift under further optimization. Low-rank adaptation (LoRA) interventions restrict updates to small trainable adapter matrices, while full fine-tuning permits all model parameters to change. The experiments vary adapter rank, brain-loss weight, loss form, and initialization to distinguish insufficient optimization capacity from absence of a specific effect.

Teacher, student, corpus, batch construction, and evaluation are held fixed within each paired real-versus-control contrast. The alignment layer is declared before scoring. The early preservation study includes a known compute caveat: the cold arm receives about twice the optimizer steps of the warm arms, so its cross-arm result is interpreted together with held-out perplexity rather than as a pure initialization effect [E003].

## 3.6 Formal interpretation and its limits

### 3.6.1 Conditional dependence and unique variance

The formal question is what dependence remains between representation $X$ and response $Y$ after nuisance variable $Z$ is known. Conditional mutual information defines that population quantity:

$$I(X;Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z). \tag{6}$$

$H(Y \mid Z)$ is the uncertainty in the response after knowing the nuisance, and $H(Y \mid X, Z)$ is the remaining uncertainty after also knowing the representation [19]. Thus Equation 6 is zero when $X$ adds no conditional dependence beyond $Z$. It licenses a statement about a fully specified joint distribution, not automatically about a finite cross-validation estimator.

Under a population linear-Gaussian model with correctly specified, unregularized conditional means, conditional mutual information is related to partial $R^2$ by

$$I(X;Y \mid Z) = -\tfrac{1}{2}\log\bigl(1 - R^2_{\text{partial}}\bigr), \qquad \Delta R^2_{\text{semi}} = \bigl(1 - R^2_{\text{reduced}}\bigr)\, R^2_{\text{partial}}. \tag{7}$$

$R^2_{\text{partial}}$ rescales the improvement by variance left unexplained by the reduced model, whereas $\Delta R^2_{\text{semi}}$ is the raw improvement in total variance explained. The equations show why both quantities express incremental dependence under the stated population assumptions. Our estimator in Equation 2 is instead finite-sample, cross-validated, and ridge-regularized. It is therefore an operational measure of incremental linear predictive value, not an estimator or lower bound for conditional mutual information.

### 3.6.2 Regularization and a conditional MAP reading

The intervention objective can sometimes be read as regularized maximum likelihood. If $\exp[-\lambda_B \mathcal{L}_B(\theta)]$ is normalizable over parameters and does not depend on the observed text sample, then it defines a prior

$$p_B(\theta) = \frac{1}{C}\exp[-\lambda_B \mathcal{L}_B(\theta)], \tag{8}$$

where $C$ is the normalization constant and $\theta$ denotes student parameters. Under those assumptions, minimizing text negative log-likelihood plus the penalty is equivalent to a maximum-a-posteriori estimate. The plain-language interpretation is that the penalty prefers some parameter settings among those that fit the text. In the implemented co-trained-target experiments, however, the brain target and readout are data-dependent, so "auxiliary loss" or "regularizer" is the primary description. Equation 8 does not prove that the preference is biologically correct or beneficial.

### 3.6.3 Data processing

The data-processing inequality answers a narrow preservation question when the required Markov chain is valid. If a student representation $C$ is constructed only by post-processing a teacher representation $T$, and brain response $B$ has no other path to $C$ once $T$ is known, then $B \to T \to C$ implies

$$I(B;C) \le I(B;T). \tag{9}$$

The inequality says that deterministic or stochastic post-processing cannot create information about $B$ that was absent from $T$ [19]. Ordinary distillation does not automatically satisfy this chain because teacher and student both receive stimulus or corpus information through side paths. Likewise, a chain such as $\theta^\star \to S \to B$ is an explicit conditional-independence model, not proof that brain responses contain no information beyond text. Equation 9 motivates an empirical preservation test but neither orders teacher and student alignment scores in ordinary distillation nor supplies a supervision budget.

### 3.6.4 Generalization information bound

The information-theoretic generalization bound asks how strongly learned parameters depend on the particular training sample. For a learning algorithm viewed as a channel from $Z^n$ to learned parameters $W$, and a $\sigma$-sub-Gaussian loss, the expected generalization gap obeys

$$\overline{\mathrm{gen}} \leq \sqrt{\frac{2\sigma^2 I(W;Z^n)}{n}}. \tag{10}$$

$n$ is the sample size and $I(W;Z^n)$ measures dependence between the output of training and the sampled dataset [20]. The right-hand side increases, rather than decreases, with this mutual information. The bound therefore does not show that adding brain information improves generalization and does not predict a benefit of order $\sqrt{\Delta I/n}$. A brain regularizer could tighten the bound only if it reduces dependence on sampled text while retaining empirical fit, which is an empirical possibility rather than a theorem.

### 3.6.5 Rate–distortion-style trade-offs

Shannon's rate–distortion function asks for the minimum information rate needed to represent a source within an expected distortion limit:

$$R(D) = \min_{P_{\hat{S}|S}:\, \mathbb{E}[d(S,\hat{S})] \leq D} I(S;\hat{S}). \tag{11}$$

$S$ is the source, $\hat{S}$ its reconstruction, $d$ a declared distortion measure, and $D$ the allowed expected distortion [19]. The thesis does not identify model parameters or FLOPs with $I(S;\hat{S})$, and it does not establish brain loss as a Shannon distortion. We use "rate–distortion-style" only as an engineering analogy for the empirical Pareto frontier among model budget, perplexity, and alignment. Equation 11 organizes the trade-off but does not predict a favorable frontier shift.

## 3.7 Experiment families

**Signal validation.** The ROI feasibility study compares trained and untrained GPT-2-family and Qwen representations under contiguous sentence folds [E002]. The conditional voxelwise confirmation uses UTS03, whole-story-held-out folds, fold-only feature reduction, a capacity-matched nuisance set, and independently selected reliable voxels [E006]. The first establishes inexpensive feasibility; the second is the load-bearing within-participant signal test.

**Quality and preservation.** Cold and warm distillation arms measure whether ordinary compression leaves alignment headroom and how alignment covaries with held-out perplexity [E003]. A separate comparison across 22 models from six families uses bits per byte as a tokenizer-robust quality measure and tests the global association, a capable-model band, and family-cluster sensitivities [E015]. These designs distinguish a broad quality relationship from an objective-specific preservation effect.

**Intervention and participant generalization.** The first intervention compares real and block-permuted averaged ROI targets across paired seeds and contiguous folds [E004][E005]. The participant-specific extension trains and evaluates targets separately for nine participants, making participant the primary inference unit [E008]. Nested and randomized participant-subset analyses test whether apparent improvement with response averaging is explained by increasing sample size or by the identities of included participants [E010]. A separate encoding-average analysis measures legitimate signal-to-noise improvement from averaging without treating that endpoint as individual-level generalization [E014].

**Alternative parameterizations.** Higher-rank LoRA tests whether the primary intervention was limited by adapter capacity [E011]. Contrastive ROI losses test whether the mean-squared-error target was the wrong geometry [E013]. A naturalistic voxelwise weight sweep tests whether the regression lever takes hold at dense spatial resolution, while gentle full fine-tuning tests whether updating all parameters changes the conclusion [E013][E017]. Each experiment addresses one mechanism boundary; none alone licenses a universal substrate null.

**Practical and privileged-information routes.** The downstream test requires a reliable brain-specific representation fulcrum before estimating out-of-domain perplexity over eight training seeds [E009]. The learning-using-privileged-information route uses gaze-derived training-only information, but gates the intervention on label predictiveness without leakage and stable paragraph-disjoint learning curves [E024]. A gate failure stops the proposed intervention rather than being reported as a completed negative training experiment.

**Stimulus-predictable and cognitive targets.** A stimulus-predictability analysis conditions on estimated response predictable from the text to ask whether a residual alignment score remains [E020]. Because the reference predictor is itself estimated and can leak or be unreliable, this experiment diagnoses a candidate ceiling rather than identifying a universal ceiling. Reading-time targets test whether a reported cognitive advantage depends on meaningful target content or can be reproduced by content-free signal shape [E021]. Reduced external reproductions of text and speech interventions are treated as corroboration or bounded non-reproduction under the implemented design, not as faithful refutations of the original studies [E019][E022].

**Synthetic targets and real-brain transfer.** The dense-target experiment distills GPT-2-medium into GPT-2 using 95,999 WikiText training sentences and 1,999 held-out sentences across six seeds [E016]. TRIBE v2 is a pretrained multimodal encoding model that maps naturalistic stimuli to predicted human fMRI responses on a standardized cortical surface [21]. For the text-only E016 stimuli, its output is a 20,484-dimensional vector of predicted cortical locations and is treated as a synthetic neural target rather than recorded fMRI. The student layer 6 hidden state is mapped to that target by the jointly trained prediction head. The textfeat control instead Gaussian-projects a frozen teacher hidden representation to the same nominal width, then uses an otherwise matched prediction head and training path. Dimension matching controls output width but not effective rank, geometry, or information, so held-out target $R^2$ is interpreted within target family rather than compared as a shared scale.

Saved students pass through a separate transfer gate on 1,000 Tuckute sentences using five contiguous nuisance-subtracted folds and student layer 7 [E016]. For each seed, the primary contrast subtracts the textfeat-over-KD real-brain gain from the TRIBE-over-KD real-brain gain. The inference unit is six training seeds evaluated on one fixed participant-averaged endpoint. The exact sign test therefore supports a statement about repeatability across training seeds for that endpoint, not participant- or population-level biological generalization.

# 4 Results

The experiments test a sequence of increasingly demanding claims. First, an alignment score must detect information that language training adds to a model representation. Second, ordinary distillation must leave enough alignment headroom for an additional objective to matter. Third, an alignment-guided objective must move held-out alignment beyond matched controls at the correct inference unit. Fourth, any movement must generalize across participants or other biologically meaningful units. Finally, it must improve an outcome that is not the alignment score itself. Table 2 summarizes where the sequence passed, failed, or stopped at a precondition.

Table 2: Overview of the experimental chain. "Unit" is the unit that licenses the stated conclusion, not the number of voxels, folds, or repeated optimization runs available in the artifact.

| Question | Records | Valid unit | Verdict and scope |
|---|---|---|---|
| Is alignment measurable beyond recorded nuisance features? | [E002][E006] | Five contiguous folds on one averaged ROI target; one participant with 20 stories arranged into five held-out-story folds | Positive trained–untrained gaps on Tuckute and conditional UTS03 evidence; no participant-population claim. |
| Does ordinary KD preserve it, and how does quality enter? | [E003][E015] | Fixed averaged benchmark; family clusters rather than individual models | Partial preservation with genuine headroom; alignment is coupled to language-model quality, but not determined by it. |
| Does a brain loss move held-out alignment? | [E004][E005] | Five fold means after averaging seeds within fold | Small averaged-target trends; no reliable near-rate gain at the valid fold unit. |
| Does the effect generalize across people? | [E008][E010][E014] | Nine participants; subset curves are sensitivity analyses | No mean per-individual gain; averaging raises SNR and changes the estimand, while the proposed dose response does not identify a causal law. |
| Do stronger or different optimization regimes rescue it? | [E011][E013][E017] | Nine participants for ROI tests; one- and three-participant feasibility probes for voxelwise tests | Extra LoRA capacity, contrastive loss, voxelwise mean-squared error, and full fine-tuning do not produce a demonstrated brain-specific gain in the tested regimes. |
| Does the signal improve a practical learning outcome? | [E009][E024] | Eight training seeds on fixed out-of-domain sets; seven paragraph clusters for the gaze gate | No detected out-of-domain-perplexity payoff when the manipulation is unreliable; the privileged-information experiment stops because its information and split preconditions fail. |
| Does a dense synthetic brain target solve the bottleneck? | [E020][E016] | One six-participant story diagnostic; six training seeds on one fixed averaged endpoint | The empirical ceiling instrument is too noisy to close the ceiling; synthetic-target fit improves, but the improvement does not transfer to the real-brain endpoint. |

## 4.1 Controlled measurability on real fMRI

The first question is whether a trained language model contains brain-predictive information beyond the recorded low-level properties of the stimulus. E002 measures this on the Tuckute sentence benchmark by fitting ridge encoding models from a fixed LM layer to five participant-averaged left-hemisphere language ROIs. The nuisance-only model contains sentence length, position, and static-embedding features, and the full model adds the contextual LM representation at matched PCA capacity. The estimand is the held-out semipartial increment, or unique $R^2$, under five contiguous stimulus folds. A same-architecture untrained network controls for feature dimension and architecture without language training [E002].

All three trained models have positive unique $R^2$, whereas every untrained seed is negative (Table 3). The trained-minus-untrained gaps are +0.030 for GPT-2, +0.033 for GPT-2-medium, and +0.050 for Qwen2.5-0.5B [E002]. The corrected mean ceiling coefficient for the five functional ROIs is 0.491 [E002][E004]. Because that coefficient and unique $R^2$ are not the same statistical quantity, their legacy ratio is not interpreted as a fraction of explainable variance. These estimates belong to one participant-averaged target evaluated over five stimulus folds; they establish conditional measurability on this benchmark, not population prevalence.

Table 3: Tuckute ROI benchmark under five contiguous folds. Unique $R^2$ is the held-out semipartial increment over the nuisance set [E002].

| Model | Layer | Trained $\Delta R^2$ (mean±SD) | Untrained $\Delta R^2$ | Gap | Source |
|---|---|---|---|---|---|
| GPT-2 | L7 | $+0.020 \pm 0.008$ | $-0.011$ | $+0.030$ | [E002] |
| GPT-2-medium | L14 | $+0.019 \pm 0.005$ | $-0.014$ | $+0.033$ | [E002] |
| Qwen2.5-0.5B | L12 | $+0.036 \pm 0.010$ | $-0.014$ | $+0.050$ | [E002] |

E006 asks whether the same trained–untrained contrast survives at voxel resolution in naturalistic data. It uses the 20-story LeBel data for participant UTS03 and selects 11,442 voxels using split-half reliability from a separate repeated story. The encoding model holds out entire stories and subtracts word and phoneme rate, duration and length, log frequency, and static-embedding nuisance features. The primary estimand is again the trained-minus-untrained unique-$R^2$ gap on the fixed UTS03 endpoint [E006].

The gap is +0.0207 for GPT-2 and +0.0277 for Qwen2.5-0.5B. Ninety-five percent and 99% of selected voxels, respectively, have positive pointwise gaps, and each model has a positive mean gap on all five held-out story blocks [E006]. The untrained unique $R^2$ is $-0.017$ in both model comparisons [E006]. These spatial and story-block signs show that the result is not confined to a small voxel set or one held-out story. They are not independent replicates: all voxels come from one participant, voxels are spatially dependent, and the story folds share training stories. The original voxel-bootstrap confidence intervals and the E006-derived E007 minimum-detectable-effect claim are therefore retracted [E006].

Table 4: Conditional voxelwise result on LeBel UTS03. Spatial prevalence and story-block signs are sensitivity checks within one participant, not population inference [E006].

| Model | Layer | Trained−untrained gap | Positive selected voxels | Positive story blocks |
|---|---|---|---|---|
| GPT-2 | L7 | +0.0207 | 95% | 5/5 |
| Qwen2.5-0.5B | L12 | +0.0277 | 99% | 5/5 |

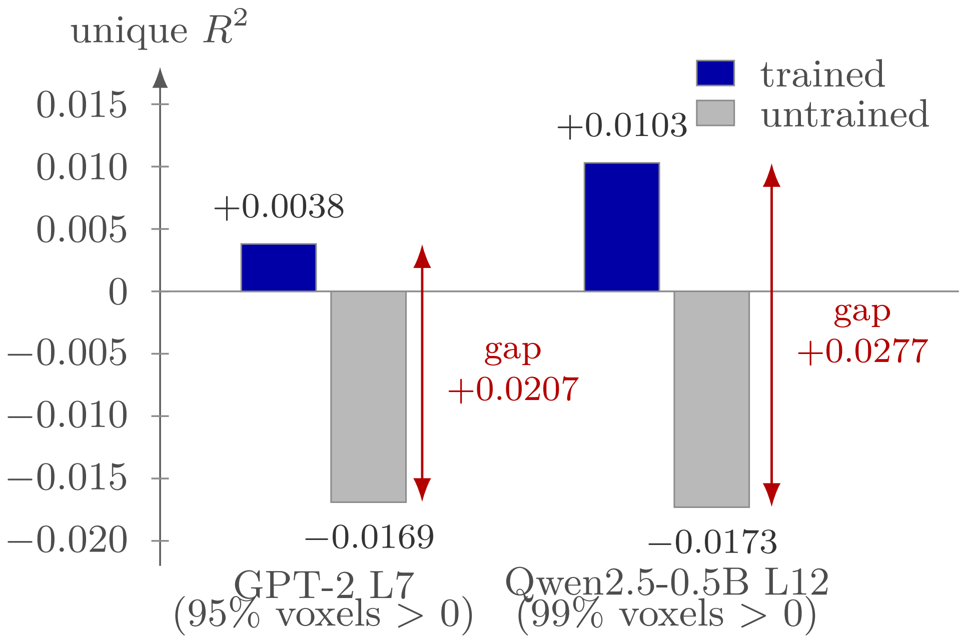


Figure 4: Mean unique $R^2$ on the selected LeBel UTS03 voxels. Arrows show the trained-minus-untrained gaps, and labels report spatial prevalence within UTS03 [E006].

E002 and E006 therefore establish the first gate: language training adds linearly accessible information that predicts held-out fMRI beyond the recorded nuisance set. They do not show that the information can be increased by training, that it generalizes across participants, or that increasing it would improve language modeling. The two datasets contribute different parts of that statement. Tuckute shows the effect on isolated sentences and a compact participant-averaged language-network target across three model families, while LeBel shows it on continuous stories with temporal controls and a dense response from one deeply sampled participant. The common trained-over-untrained direction makes the measurement premise less dependent on a single stimulus format. The different biological units prevent their combination into one population effect, so they are treated as complementary existence evidence rather than a meta-analysis.

### 4.2 Distillation leaves headroom, but alignment is coupled to model quality

E003 asks whether ordinary logit knowledge distillation preserves the teacher's alignment without an explicit brain objective. The load-bearing arm initializes a GPT-2 student randomly and trains it from the logits of GPT-2-medium. Warm-start KD, ordinary language-model fine-tuning, pretrained GPT-2, distilgpt2, and an untrained network provide reference points. All arms are scored at fixed layers on the same participant-averaged Tuckute target and nuisance-controlled contiguous folds [E003].

The cold-init student ends +0.0181 below its teacher in unique $R^2$ ($p < 0.001$ under the recorded bootstrap) and retains 0.37 [0.14, 0.58] of the teacher–floor interval [E003]. This rules out preservation for free in the tested from-scratch run and creates measurable headroom. The magnitude is not an objective-specific estimate because the cold student remains under-trained: held-out perplexity is 477, compared with 74 for its teacher and 44–95 for the converged warm arms. Absolute unique-$R^2$ values also vary with PCA rank, although the retention ordering survives $n_{\text{PCA}} \in \{25, 50, 100\}$ [E003].

Table 5: Distillation and reference arms on the participant-averaged Tuckute benchmark [E003]. Floor-anchored retention is $\rho' = (A_S - A_0)/(A_T - A_0)$; distilgpt2 uses GPT-2 as its lineage teacher.

| Arm | Initialization/objective | $\Delta R^2$ | $\rho'$ [bootstrap CI] | Held-out ppl | Interpretation |
|---|---|---|---|---|---|
| GPT-2-medium | pretrained teacher | +0.0188 | — | 74 | teacher reference [E003] |
| GPT-2 | pretrained | +0.0195 | 1.02 [0.77, 1.27] | 105 | inherited alignment [E003] |
| LMFT-warm | warm, language loss | +0.0174 | 0.95 [0.81, 1.13] | 44 | drift control [E003] |
| KD-warm | warm, logit KD | +0.0144 | 0.84 [0.67, 1.04] | 95 | drift under KD [E003] |
| distilgpt2 | released distilled model | +0.0076 | 0.60 [0.42, 0.84] | 169 | lineage reference [E003] |
| KD-cold | random, logit KD | +0.0005 | 0.37 [0.14, 0.58] | 477 | under-trained headroom |
| untrained | random floor | −0.0102 | 0 | $\sim 6 \times 10^4$ | floor [E003] |

The same run exposes the quality confound. Across its five non-teacher trained/reference arms, alignment correlates with log perplexity at Pearson −0.88 [E003]. That correlation uses few heterogeneous points and is not a general law, so E015 repeats the analysis with a tokenizer-independent quality measure across 22 models from six decoder families. Its primary full-range association between mid-layer unique $R^2$ and log bits per byte is $r = -0.783$, with family-cluster bootstrap 95% interval $[-0.911, -0.500]$ [E015]. Leave-one-family-out estimates remain between −0.738 and −0.833 [E015]. The 22 models are not 22 independent scientific replicates: the six families form approximately two model generations, and models within scaling families share training recipes.

The coupling weakens where current brain-tuning comparisons operate. For models with bits per byte below 1.30, $r = -0.501$ with Fisher interval $[-0.859, +0.188]$; in the overlap band $[1.13, 1.26]$, $r = -0.480$ with interval $[-0.885, +0.340]$ [E015]. Those capable-band intervals include zero. A scale-adjusted partial correlation remains negative at −0.675 with Fisher interval $[-0.857, -0.344]$, but bits per byte and parameter count are highly collinear, so this does not isolate a pure quality mechanism [E015]. Family residuals are also not identified because the families lack common quality support; the recorded ANCOVA family test has $p = 0.204$ [E015].

The combined conclusion is narrower than either "KD destroys brain information" or "alignment is just perplexity." Ordinary KD can leave a student below its teacher, but E003 does not distinguish objective-specific shedding from under-training and quality. E015 shows that quality is a necessary matching variable across a broad modern-decoder comparison, while its capable-band and architecture-residual analyses show that output quality does not determine the hidden representation. The intervention experiments therefore compare real brain targets with matched-regime or matched-perplexity controls rather than a vanilla model alone. This distinction also changes the interpretation of "headroom." The teacher–student gap is an opportunity only if some of it is caused by an internal choice that an auxiliary objective can alter without paying the same cost in perplexity. E003 establishes the gap but not that counterfactual. E015 makes unequal quality a credible alternative explanation, yet its capable-band uncertainty leaves room for representations of similar output quality to differ in alignment. The intervention stage is needed precisely because neither preservation nor cross-model correlation identifies the movable component.

### 4.3 The apparent intervention gain depends on the inference target

E004 is the first direct lever test. It co-trains a Qwen2.5-0.5B student with an MSE brain term and compares it with an otherwise identical block-permuted-target twin [E004]. Stimulus folds, optimization budget, nuisance model, student, teacher, and brain-loss weight are held fixed. The estimand is the real-minus-permuted change in held-out unique $R^2$ relative to the same untuned base [E004].

The point estimate is +0.00316. The original interval treated three seeds by five folds as 15 independent observations and excluded zero. After averaging seeds within fold, the valid five-

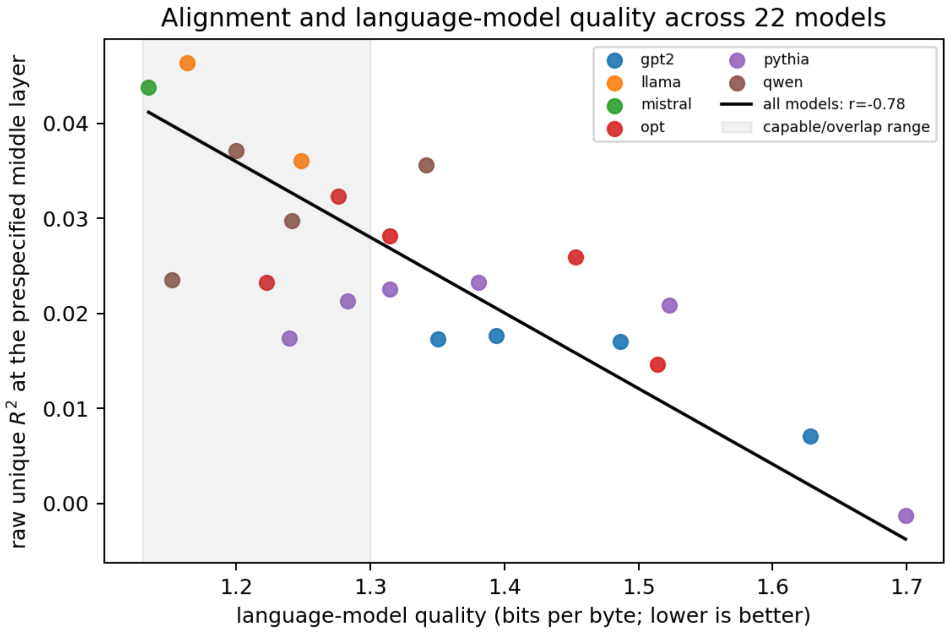


Figure 5: Tokenizer-independent language-model quality and controlled mid-layer alignment across the 22 E015 models. The fitted line summarizes the full range; the shaded region marks the capable/overlap range in which the association is weaker and individually uncertain. Points share family and generation structure, so they are not 22 independent scientific replications [E015].

fold $t$ interval is $[-0.00227, +0.00858]$ [E004]. Fold 4 supplies 64% of the summed contrast, and removing it reduces the estimate to +0.0014. The run's own unpaired permutation exceedance flag is false. E004 therefore provides no inference-unit-valid evidence of a held-out lever, while its five folds cannot exclude a small effect near the observed mean.

E005 asks the related matched-perplexity question against the five-participant-averaged target. At $\lambda = 10$, its real-minus-permuted mean is +0.0081. The flat seed-by-fold bootstrap interval $[+0.0023, +0.0171]$ is pseudo-replicated; the five-fold $t$ interval is $[-0.0030, +0.0193]$, the median is +0.0034, and one fold supplies 52% of the mean [E005]. Four of five fold means are positive, so the result is a small averaged-target trend rather than a precise null.

The later $\lambda$ sweep separates a near-rate improvement from a rate–distortion trade. At $\lambda = 1$, the real-minus-matched-permuted contrast is +0.0030 with fold interval $[-0.0025, +0.0085]$ [E005]. At $\lambda = 3$, it is +0.0063 with interval $[-0.0092, +0.0218]$, and at $\lambda = 10$ it is +0.0072 with interval $[-0.0044, +0.0188]$ [E005]. At $\lambda = 30$, the contrast is +0.0060 $[+0.0002, +0.0119]$, but held-out perplexity worsens from approximately 51.5 for KD-only to 67.4 [E005]. The only fold-level-positive arm therefore buys alignment by moving to a worse language-model operating point, not by providing a free matched-rate improvement.

E008 changes the biological inference unit. It repeats the real-versus-permuted comparison separately for nine participants, using three optimization seeds and five shared folds within each participant. The estimand is the across-participant mean of each participant's seed-and-fold-aggregated real-minus-permuted effect. The mean is +0.00010, the nine-participant $t$ interval is $[-0.00037, +0.00058]$, and 5/9 participant effects are positive (two-sided sign $p = 0.50$) [E008]. The fold-clustered sensitivity estimate is +0.00026 with interval $[-0.0004, +0.0009]$, and every leave-one-fold-out analysis fails to establish a positive effect. The observed participant variance gives an 80%-power minimum detectable effect of approximately +0.0006 to +0.0013, well below the averaged-target estimate [E008].

The four participants used to construct the earlier averaged target have an individual mean

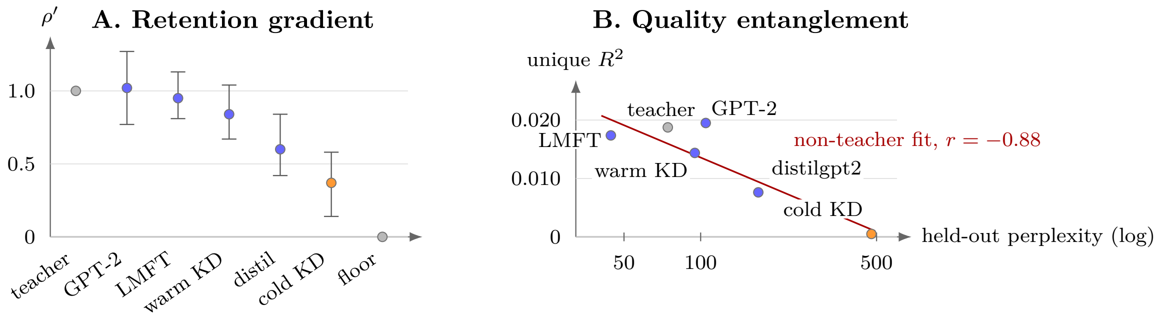


Figure 6: Plain KD leaves alignment headroom, but alignment co-varies with language-model quality in the E003 reference set. The line and coefficient describe the five non-teacher E003 points, not the corrected cross-family E015 law [E003].

of $-0.00010$ [E008]. The five held-out participants have mean $+0.00026$, also with an interval containing zero [E008]. Thus, the E005 estimate is not an estimate of a mean per-person effect. It is an effect on a different, higher-SNR target: the response shared across the averaged participants.

E010 tests whether participant count alone causes a smooth increase in the intervention gap. Its initial nested subsets give gaps of $-0.0002$, $-0.0001$, $+0.0023$, $+0.0194$, and $+0.0071$ for $k = 1, 2, 3, 5, 9$ averaged participants [E010]. Because subject identity changes with $k$, this design confounds count with which participants enter the average. The random-subset rerun gives $+0.0015$, $-0.0013$, $+0.0088$, and $+0.0094$ for $k = 1, 3, 5, 9$, with standard deviations between 0.005 and 0.015 [E010]. The non-monotone, noisy rerun does not identify a causal dose-response law. It remains suggestive corroboration that averaged and individual targets behave differently.

E014 isolates the measurement effect without training an LM. Using the same Qwen representation, it compares encoding of individual targets with encoding of random participant averages. Mean per-participant unique $R^2$ is $+0.0069$, with 7/9 participants positive, whereas the standard five-participant average scores $+0.0357$ [E014]. The count sequence for $k = 1, 2, 3, 5, 9$ is $+0.0039, +0.0162, +0.0324, +0.0444, +0.0702$ [E014]. Because individual scores are already positive, this is not evidence that averaging fabricates an otherwise absent encoding signal. It shows legitimate SNR improvement and a change from individual-response alignment to alignment with the shared response component.

Taken together, E004 and E005 provide small averaged-target trends but no reliable near-rate lever at five-fold inference. E008 supplies the load-bearing biological result: no mean per-individual gain in nine participants under the tested matched-regime objective. E010 does not convert the averaged-versus-individual contrast into a clean causal dose response, and E014 shows why a larger score on an averaged target can still be a legitimate measurement of a different estimand. The apparent contradiction is therefore resolved at the level of the question being estimated. E005 asks whether training helps predict one denoised group-average response, and its small positive trend is compatible with an intervention that follows highly shared stimulus structure. E008 asks whether the same real-versus-control advantage is positive on average across individual people, and it finds a tightly bounded mean near zero for the predeclared effect size. Neither result invalidates the other measurement, but only the latter supports inference to participant variation. This is why response averaging is described as an estimand change rather than as a universally invalid artifact.

## 4.4 Alternative intervention regimes do not rescue the tested lever

E011 asks whether the E008 result is caused by insufficient low-rank adaptation (LoRA) capacity. It increases adapter rank from 16 to 64 and doubles the training duration while holding the target, objective, participant set, and matched-regime control fixed. Across the same nine participant units, the mean real-minus-permuted effect is $+0.00043$ with $t$ interval $[-0.0005, +0.0013]$ [E011]. Six of nine effects are positive, but one participant contributes 79% of their sum; leaving that participant out returns the estimate to approximately $+0.00010$ [E011]. Absolute alignment and perplexity move about as much as in E008, so this result establishes robustness to extra adapter capacity at the same effective operating point, not robustness to every stronger manipulation.

E013b changes the loss geometry from MSE to symmetric contrastive training while keeping the nine-participant ROI design. Its mean is $-0.00019$ with participant $t$ interval $[-0.0008, +0.0005]$, 4/9 positive signs, and a matched-perplexity intercept of approximately $-0.0003$ [E013]. The contrastive objective therefore does not rescue the ROI result. The scope is limited by the five-dimensional ROI target, which is a weaker test of a contrastive objective than a high-dimensional voxel target.

The voxelwise E013 route tests whether the MSE distillation mechanism can first create an improving representation on naturalistic LeBel data. Without a KD anchor, real-target tuning reduces held-out alignment below the untuned base in all six subject-by-seed runs, so the first run is a failed manipulation rather than a null effect [E013]. With a KD anchor on UTS03, the untuned base is approximately 0.0187 unique $R^2$ [E013]. At $\lambda = 1$, real-target tuning reaches 0.0183 with real-minus-permuted gap $-0.0009$; at $\lambda = 3$ and 6, it falls to 0.0086 and 0.0130, with gaps $-0.0049$ and $-0.0039$ [E013]. No tested weight improves over the base or beats its permuted twin. This is a single-participant mechanism failure for voxelwise MSE distillation, not a population null for every voxelwise objective.

E017 changes the parameterization to full fine-tuning. An aggressive configuration produces no manipulation success in three UTS03 seeds and increases perplexity by a factor of 1.63 [E017]. A gentler configuration preserves perplexity more closely and is then run for three seeds on each of UTS01, UTS02, and UTS03. Across the nine subject-by-seed cells, the recorded real-minus-permuted mean is $+0.0003$, with descriptive cell-level $t$ interval $[-0.0002, +0.0008]$ and $p = 0.274$ [E017]. Per-subject means are $+0.0005$, $-0.0001$, and $+0.0006$ [E017]. The valid biological unit is the participant, so the nine-cell interval is not population inference; the run is a three-participant feasibility result showing no demonstrated brain-specific gain in this configuration.

These variants close specific explanations rather than the entire hypothesis space. The ROI result is not rescued by more adapter capacity or a contrastive loss. The tested voxelwise MSE mechanism fails to improve the base representation, and the tested full-fine-tuning configurations either damage language quality or remain indistinguishable from their permuted twins. Different data, targets, objectives, or substantially larger participant cohorts remain outside these results. The sequence nevertheless removes several immediate rescue explanations. If the primary null were caused only by insufficient low-rank capacity, E011 should have enlarged the contrast; if squared-error geometry were uniquely unsuitable for ROI targets, E013b should have improved it; if adapters were the bottleneck, the gentle full-fine-tuning regime in E017 should have separated real and control targets. None does so under its valid scope. The voxelwise branch is even earlier in the causal chain because stronger weights fail the manipulation check itself. These results therefore narrow the plausible success regime toward different data or target construction rather than a simple increase of the same optimization knob.

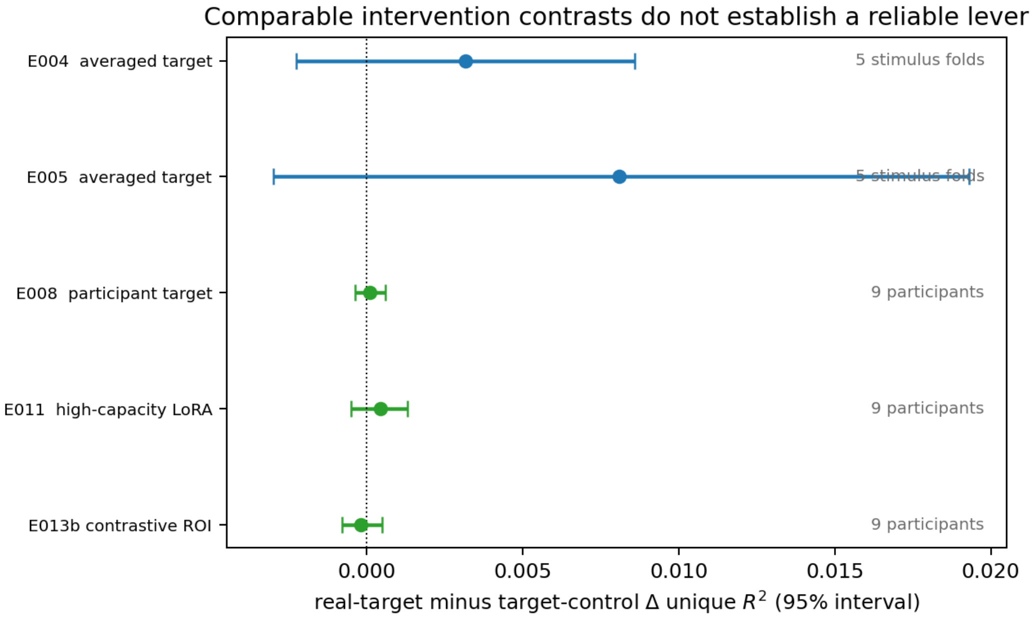


Figure 7: Comparable real-target minus target-control intervention contrasts. E004 and E005 use five stimulus folds; E008, E011, and E013b use nine participants. The rows are displayed together to show scale and uncertainty but are not pooled because their estimands and inference units differ [E004,E005,E008,E011,E013].

### 4.5 No practical payoff is demonstrated when the prerequisite signal does not move

E009 asks whether alignment-guided KD improves a non-brain outcome: out-of-domain perplexity on Pereira and LeBel text. It compares KD-only, real-brain mean-squared error (MSE), block-permuted MSE, and a text-feature auxiliary target over eight training seeds. The mediator that would make the downstream test interpretable is the real-minus-permuted alignment change on the in-domain averaged target [E009].

That mediator has mean +0.011, median +0.0042, and 5/8 positive seed signs. Its measured 80%-power minimum detectable effect is 0.023, and seed 0 contributes +0.063 while the remaining seven seeds average approximately +0.004 [E009]. The representation fulcrum is therefore not reliable. Every OOD-perplexity contrast also lies inside its measured minimum detectable effect: real MSE minus KD-only is +0.009 on Pereira and +0.053 on LeBel, while real MSE minus the permuted twin is −0.019 and −0.068 [E009]. E009 supports no detected brain-specific out-of-domain (OOD) perplexity payoff under this all-data, averaged-target setup. It is a bounded result because the manipulation whose payoff was being tested did not reliably take hold, and because robustness and sample-efficiency outcomes were not run.

E024 tests whether a cognitive measurement can help through a different channel: learning using privileged information (LUPI), where the auxiliary measurement is available only during training. The planned five-arm ZuCo intervention is gated on two preconditions: natural-reading gaze must contain information about the target label, and paragraph-disjoint splits must support a stable low-sample learning curve. The data comprise 300 natural-reading sentences from 12 participants, but only seven source paragraphs [E024].

The gaze measurement itself is reliable: combined split-half reliability is 0.71 [E024]. It is not informative for the target under natural reading. A rich 35-dimensional gaze probe gives binary AUC 0.508 with permutation $p = 0.45$, and five-way relation-type balanced accuracy is 0.202 against 0.200 chance, $p = 0.35$ [E024]. Linear, nonlinear, participant-stacked, and within-participant probes give the same null pattern. Under task-directed reading, the relation-type probe reaches 0.295 against 0.100 chance, with permutation $p = 0.003$ [E024]. This contrast shows that gaze contains relation information when readers are instructed to search for the relation, but that channel is task-intent leakage rather than portable natural-reading privileged information [E024].

The seven-paragraph structure independently fails the split gate. Across paragraph-disjoint splits, test size ranges from 8 to 161 sentences and the empirical low-$n$ minimum detectable effect ranges from approximately 0.03 to 0.08 or becomes undefined [E024]. No stable leakage-free sample-efficiency estimand is available on this substrate. The binding gate therefore stops the five-arm intervention before training. This is evidence that the chosen privileged signal and dataset fail the required preconditions, not evidence that learning using privileged information is universally ineffective. E009 and E024 fail at different locations and should not be collapsed into a single downstream null. E009 implements downstream evaluation but lacks a reliable brain-specific mediator, so it cannot reveal what a successful alignment change would do. E024 never reaches the intervention because the clean natural-reading privileged signal is neither label-informative nor stably evaluable under paragraph-disjoint splits. Together they show why practical claims require a verified chain from target information to representation movement to held-out task performance. They do not establish that every neural or cognitive signal lacks practical value.

### 4.6 Synthetic-target improvement does not transfer to the common real-brain endpoint

E020 first asks whether the part of fMRI not shared across participants contains alignment with a common stimulus representation. For denizenslab story 11, it estimates the shared response with a leave-one-participant-out mean across six participants and defines each participant's residual relative to that estimate. The fixed Qwen2.5-0.5B representation is evaluated with linear ridge encoding on higher-language ROIs [E020]. The primary quantities are raw per-vertex correlations with the shared response, total response, and residual, all on one scale [E020].

The initial correlations are $A_{\text{shared}} = +0.176$, $A_{\text{total}} = +0.108$, and $A_{\text{resid}} = +0.090$ [E020]. The residual positive is not a valid non-stimulus signal. The leave-one-out reference reliability is only 0.33, residual split-half reliability is 0.174, and the residual estimate remains stimulus-contaminated as the reference grows from one to five participants [E020]. After gapping folds for hemodynamic autocorrelation and partialling temporally aligned stimulus structure, the trained-minus-untrained residual gap is $-0.018$ [E020]. This diagnostic finds no surviving non-stimulus residual alignment on the fixed story, but the noisy shared-response reference cannot quantify a tight ceiling. E020 is therefore a failed ceiling instrument with a successful artifact guard: bounded, not closed.

E016 then asks a different question: whether a dense stimulus-derived brain target can be optimized at scale and transfer to measured fMRI alignment. TRIBE v2 is a pretrained multimodal model that predicts high-resolution cortical fMRI responses for new stimuli [21]. E016 uses its 20,484 predicted cortical values per text stimulus as a synthetic neural target, not as newly recorded fMRI. TRIBE and a projected teacher-hidden-state control each define their own synthetic target; KD-only and block-permuted variants provide within-target references. The two synthetic target spaces are dimension-matched in the training harness but differ in rank, geometry, and baseline difficulty, so their raw target-$R^2$ values are not directly commensurate [E016].

Across six training seeds, TRIBE-target training improves its held-out synthetic target $R^2$ over KD-only by +0.078298. The projected text-feature target improves its own endpoint by +0.000903 [E016]. The maximum relative perplexity deviation across the checked target and control arms is 0.009220 [E016]. The supported synthetic claim is therefore that TRIBE beats this particular sentence-local projected teacher-hidden-state control on their respective

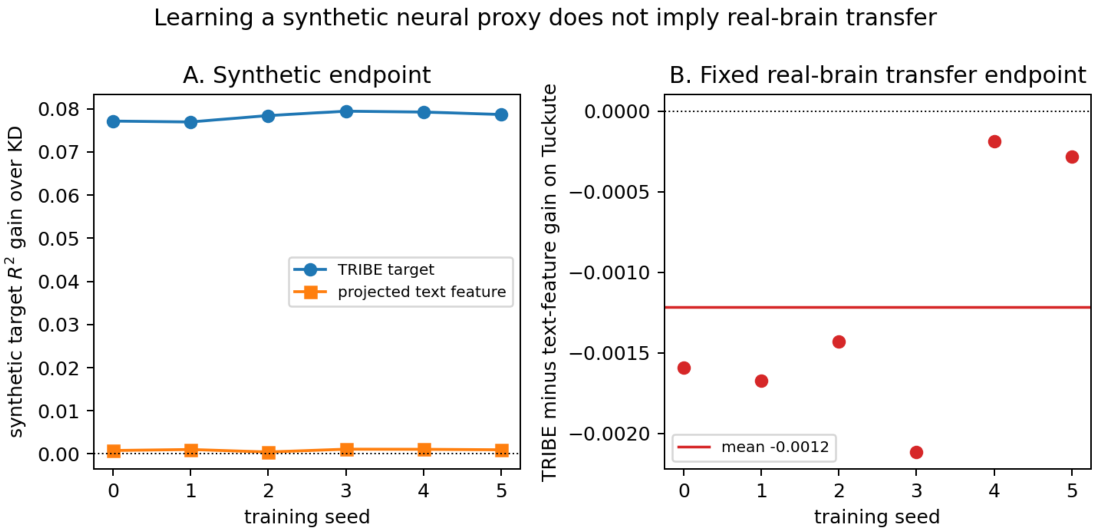


Figure 8: E016 separates proxy optimization from biological transfer. Panel A shows each seed's gain on its respective synthetic endpoint; the projected text-feature control is dimension-matched but not information-matched. Panel B shows the TRIBE-minus-text-feature margin on one common Tuckute endpoint, with all six margins below zero. The seeds quantify training variability for this fixed averaged target, not participant or population uncertainty [E016].

endpoints under nearly unchanged language quality. It does not establish superiority over matched-information controls, non-stimulus information, or brain specificity.

The saved students are then scored on one common Tuckute endpoint. Synthetic training uses layer 6, while the real-brain evaluation uses layer 7 and the same five-participant/five-ROI average for every seed [E016]. Relative to KD-only, the TRIBE-minus-text-feature difference is $-0.001213$; all six seed margins are negative, giving exact sign $p = 0.03125$. The corresponding difference relative to the two block-permuted gains is $-0.001005$, with five of six seed margins negative [E016]. The block permutation leaves 0–30% of rows fixed and duplicates or omits one row in five of six seeds, so that second contrast is a sensitivity analysis rather than a load-bearing exact-null comparison.

The decisive E016 result is a failure of transfer across endpoints. The synthetic target is learnable, but its improvement does not produce improved alignment on the fixed real-brain benchmark. The inference unit is six stochastic training seeds over one endpoint that already averages five participants and five ROIs. It supports neither participant-level nor population-level inference, and it does not show that all synthetic brain targets must fail. Together with E020, it locates the remaining bottleneck after measurement: constructing a dense stimulus-derived target is not sufficient unless target-specific training gains survive evaluation on recorded brain responses. E020 and E016 also separate two meanings of a ceiling. E020 tries to estimate the stimulus-predictable component of recorded fMRI and fails because the empirical reference is too noisy and leakage-sensitive to support a universal residual bound. E016 instead creates a target that is dense and learnable by construction, then asks an operational transfer question. Its negative common-endpoint margin is stronger than a failure to fit the proxy because the proxy fit succeeds in every seed. It remains narrower than a biological ceiling because the transfer endpoint is fixed, participant-averaged, and evaluated at a different student layer.

# 5 Discussion

## 5.1 Main empirical answer

The experiments separate the existence of a measurable correspondence from its value as an intervention target. Under the recorded controls, trained middle-layer representations predict brain responses beyond the low-level nuisance set and same-architecture untrained controls [E002] [E006]. Ordinary distillation leaves alignment headroom below the teacher, but that headroom is entangled with language-model quality rather than identified as an objective-specific loss [E003] [E015]. When neural targets enter the training objective, the tested methods do not establish a reliable participant-level gain or a practical downstream benefit [E004] [E008] [E009] [E011] [E013] [E017]. The answer is therefore bounded but coherent: brain alignment is measurable in the studied settings, yet it is not a demonstrated reliable training signal for the tested distillation regimes.

This conclusion is stronger than a failure to obtain a positive score and narrower than a universal impossibility claim. The program tests each link required for brain-guided distillation: valid measurement, preservation, controllable movement, participant generalization, and downstream transfer, each with a control suited to that link. Several apparent positives weaken at precisely those transitions. The value of the evidence lies in locating those failures rather than combining unlike endpoints into one verdict.

The result also changes what should count as success in this area. An intervention should not be credited merely because its optimized target rises, because an auxiliary task can become easier through generic regularization or target geometry. It should not be credited merely because a confidence interval over seeds excludes zero, because optimizer repeatability is not biological generalization. It should not be credited merely because perplexity remains acceptable, because similar output quality can coexist with different hidden representations. The complete success condition is a target-specific movement at the correct inference unit that transfers to an endpoint independent of the training proxy.

## 5.2 Where the causal chain breaks

The first break is between measurement and control. A linear readout can detect brain-relevant distinctions in a frozen representation even when a small auxiliary loss cannot reorganize that representation reliably during distillation. The neural term competes with a much larger language objective, the target is noisy and low-sample, and changing the representation can degrade language quality before it creates a stable target-specific gain. The fold-level E004 result, the participant-level E008 result, and the neutral-to-harmful voxelwise trajectory in E013 each locate a failure at this intervention stage under a different readout or inference unit [E004] [E008] [E013].

Increasing flexibility does not by itself resolve the problem. Higher-rank LoRA changes capacity without producing a stronger effective manipulation, a contrastive ROI objective does not rescue the participant-level effect, and full fine-tuning alternates between negligible movement and catastrophic forgetting before converging on a near-zero brain-specific contrast [E011] [E013] [E017]. These results implicate the tested induction mechanisms rather than one adapter rank or one loss parameterization. They do not rule out objectives with different targets, architectures, temporal models, or optimization regimes.

The second break is between optimizing a proxy and transferring to biology. E016 shows that a dense brain-derived synthetic target can become substantially easier for a student to predict under nearly unchanged language quality, while the same students do not improve on a common real-brain endpoint [E016]. Synthetic learnability is therefore evidence that an optimization channel exists, not evidence that the optimized geometry is brain-specific or biologically transferable. A training claim requires both a target-specific control and an independent real-brain transfer test.

### 5.3 Inference unit and response averaging

The inference unit changes the scientific answer because different repetitions support different claims. Pooling three seeds over five shared stimulus folds produced an apparently precise E004 interval, but the valid fold-level analysis has five units and includes zero [E004]. Likewise, the participant-averaged target in E005 supports an apparent gain, whereas the corresponding effects across nine individually evaluated participants center near zero in E008 [E005] [E008]. Seeds measure optimization variability, folds measure stimulus variability, and participants measure biological variability; one axis cannot substitute for another.

Response averaging is not merely a biased estimate of the average participant. It suppresses idiosyncratic noise and emphasizes the component evoked consistently by the shared stimulus, thereby increasing reliability and changing the estimand. The nested E010 curve suggested a dose response, but that pattern did not survive random-subset reanalysis, so it cannot carry a causal averaging law [E010]. E014 independently shows that averaging can legitimately increase an encoding score when individual signals are already positive, through higher signal-to-noise ratio and a shift toward the shared component [E014]. The defensible conclusion is thus specific: the positive averaged-target intervention result does not generalize to a mean per-individual effect in this cohort, while averaging itself is not generally an artifact.

### 5.4 What the negative results establish

The negative findings constrain particular mechanisms at particular inference units. E013 establishes that the tested voxelwise MSE distillation lever does not take hold across its weight range; it does not establish that naturalistic voxelwise objectives are impossible [E013]. E009 detects no out-of-domain perplexity benefit in a regime where the prerequisite representation contrast is unreliable, so it does not estimate the payoff of a successful alignment manipulation [E009]. E024 stops before a five-arm privileged-information intervention because natural-reading gaze is uninformative about the task label and the paragraph structure cannot support the intended learning curve; it is a failed precondition, not a completed LUPI null [E024]. E020 finds that an apparent participant-residual signal is explained by stimulus leakage and autocorrelation under its strongest diagnostic, but its noisy reference prevents the experiment from closing a universal stimulus-predictability ceiling [E020].

Three secondary experiments extend these boundaries without becoming external refutations. The faithful-as-feasible Negi-style head in E019 produces no positive brain-specific gain: gentle tuning barely moves the model, whereas stronger tuning damages both perplexity and encoding [E019]. Differences in architecture, dataset, training duration, and metric prevent that result from adjudicating the original Negi study; it instead corroborates the mechanism failure observed in the primary experiments. In E021, a phase-randomized target that preserves the reading-time signal's autocorrelation and marginal shape reproduces its auxiliary-regularization advantage, so the gain is attributable to signal shape rather than cognitive content in that setup [E021]. In E022, a reduced single-participant speech experiment does not reproduce the predeclared phoneme benefit, making the planned matched-quality follow-up vacuous, but the reduced dataset and protocol do not refute the multi-participant Moussa result [E022].

### 5.5 Requirements for future brain-guided training

The evidence implies a stricter sequence for future intervention claims. First, the alignment metric must survive leakage controls, a capacity-fair nuisance model, reliability selection independent of the test set, and an untrained-model comparison. Second, the training manipulation must beat a target-specific twin while preserving or explicitly matching language quality. Third, the effect must be evaluated at the inference unit named by the claim, with participants required for biological generalization. Fourth, a synthetic or averaged target must transfer to a separately measured real-brain endpoint. Only after these gates hold is a downstream comparison interpretable as the payoff of alignment rather than as generic regularization, language-quality change, or target learnability.

The requirements also clarify what kind of neural target may be worth pursuing. A larger or denser target is useful only if its additional dimensions encode transferable structure rather than making an auxiliary task easier. Privileged information is useful only if it predicts the target concept without importing task-induced leakage. More participants can improve both reliability and the scope of inference, but averaging them into one response cannot replace participant-level evaluation when the claim concerns people. These are identification requirements, not preferences about model architecture.

The sequence suggests an efficient future workflow as well as a stricter one. Before expensive training, a candidate neural or cognitive target should pass reliability, label-relevance, leakage, and split-stability gates. During training, manipulation success should be evaluated separately from outcome success, so a failed objective is not mistaken for evidence that a useful representation change has no consequence. After training, the real target and its control should be scored on a common endpoint with stable model identity and matched preprocessing. This ordering prevents compute from being spent on a downstream claim whose mediator or inference unit was never established.

### 5.6 Relation to prior positive studies

Prior positive studies and the present negative results address overlapping but non-identical estimands. Brain-guided fine-tuning can begin from an already capable representation, use larger neural datasets, or evaluate a modality-specific downstream task, whereas the present thesis emphasizes compression, matched quality, and participant-aware inference [8–11, 14]. The findings therefore do not show that neural supervision is never useful. They show that a gain over an untuned baseline does not, by itself, identify a brain-specific training effect and that positive proxy fit does not guarantee biological transfer.

The most durable synthesis is methodological rather than adversarial. Measurement, intervention, generalization, and payoff are separate estimands with separate controls. Results that satisfy one stage remain valuable even when a later stage fails, but they should not inherit the stronger interpretation of that later stage. This distinction allows positive and negative studies to be compared at the level of their actual interventions and inference units rather than by headline score alone.

## 6 Limitations

### 6.1 Biological generalization

The voxelwise measurability result is conditional on one deeply sampled participant, UTS03. Spatially dependent voxels increase resolution but do not provide population replication, and the five story folds share training stories [E006]. The primary intervention generalization test includes nine participants exposed to the same 1000 sentences, so participant diversity and stimulus diversity are both limited [E008]. The three-participant LeBel intervention variants are existence probes rather than population estimates [E013] [E017] [E019]. These limitations are asymmetric: a controlled within-participant positive can establish that a signal exists under that participant's measurement process, while a within-participant null cannot establish absence across the population. A decisive population claim would require more independently sampled people, prespecified participant-level aggregation, and an effect large enough to survive between-person heterogeneity.

## 6.2 Stimulus and target coverage

The Tuckute benchmark contains isolated English sentences summarized into five left-hemisphere language ROIs, whereas the LeBel experiments use naturalistic English stories from a small number of deeply sampled listeners. Neither substrate represents the full range of linguistic contexts, modalities, languages, or cortical systems. The isolated-sentence benchmark favors sentence-local representations and removes discourse history, whereas story listening introduces temporal context, acoustic structure, and hemodynamic alignment. Agreement across them is useful robustness evidence, but disagreement would not by itself identify whether stimulus format, modality, target resolution, or participant sampling caused the difference. The gaze experiment uses only 300 natural-reading sentences from seven highly unequal paragraphs, which prevents a stable paragraph-disjoint sample-efficiency curve [E024].

## 6.3 Construct validity

Unique $R^2$ is conditional on the nuisance variables actually included. The pipeline subtracts rate, duration, frequency, and static lexical or semantic features, but it deliberately leaves surprisal and imageability outside the nuisance set; "beyond nuisance" therefore does not mean beyond all stimulus-predictable content [E006]. The trained-minus-untrained gap identifies what language training adds over a random network under those controls, not shared cognitive mechanism or representational identity. E020 further shows that estimating a stimulus-predictability ceiling is limited by the reliability of the empirical reference [E020].

## 6.4 Intervention strength

The tested objectives cover MSE, a contrastive ROI loss, LoRA capacity changes, full fine-tuning, behavioral auxiliary targets, and one dense synthetic target, but they do not span all possible neural objectives or optimization schedules. In E011, additional adapter capacity does not produce a materially stronger representation change, while E013 and E017 show that more aggressive intervention can instead damage the model [E011] [E013] [E017]. A failed manipulation cannot identify the consequence of a successful one. Conversely, a manipulation that strongly fits its training target can still change the wrong internal degrees of freedom, as the synthetic-to-real transfer gate demonstrates. Future objectives need both a manipulation check on the optimized target and an independent biological endpoint that was not used to construct that target.

## 6.5 Inference and power

E004 has five valid seed-averaged fold units, and its corrected interval cannot exclude a small effect near the observed mean [E004]. Several sensitivity analyses reuse folds, stories, or participants and are therefore not independent replications. The block-permutation implementation can leave some target rows fixed, so it is treated as a sensitivity control rather than an exact marginal-preserving randomization null [E016]. Minimum-detectable-effect calculations bound the tested contrasts but do not convert a null into evidence of exact zero. The strongest null in the thesis is E008 because its participant-level interval excludes the predeclared +0.003 effect under the tested ROI regime; smaller effects and effects in other regimes remain possible [E008]. Other nulls are correspondingly narrower when their valid unit count is smaller or when the intervention itself fails to move the representation.

## 6.6 External reproduction

E019 changes the architecture, dataset, training duration, and some implementation details relative to Negi, so it provides corroboration on the available substrate rather than a faithful refutation [E019]. E022 uses one participant and a reduced speech setup rather than the full multi-participant Moussa protocol, so its non-reproduction is likewise local [E022]. These experiments test whether the reported mechanisms transfer into the available controlled pipeline; they do not estimate the original studies' effects.

### 6.7 Synthetic-target comparability

TRIBE and textfeat have the same nominal output width but different rank, geometry, and baseline difficulty. Their held-out target $R^2$ values are therefore meaningful within target family, not as a shared cross-family scale [E016]. The transfer endpoint averages five participants and five ROIs, and the synthetic objective and transfer readout use different student layers. Six stochastic seeds over this one fixed endpoint support a training-seed diagnostic, not participant- or population-level inference.

### 6.8 Downstream-task coverage

The primary practical test measures out-of-domain perplexity rather than task accuracy, robustness, calibration, or sample efficiency across a broad benchmark suite [E009]. Because its alignment manipulation is unreliable, the test does not identify the downstream value of a successful brain-specific representation change. The gaze-privileged route stops before intervention and the reduced speech route fails its reproduction gate, leaving downstream utility bounded by preconditions rather than exhaustively tested [E022] [E024].

## 7 Conclusion

This thesis tested a transition that brain-alignment research often leaves implicit: from using a linear encoding score to describe a trained model to using the same kind of signal to shape a smaller model during knowledge distillation. The transition requires more than a positive brain score. The score must survive controls for nuisance structure and leakage, ordinary distillation must leave meaningful headroom, a neural objective must move held-out alignment beyond a target-specific control at comparable language quality, the effect must generalize at the intended biological inference unit, and the resulting change must transfer to an independent neural or practical endpoint. The experimental program evaluated those requirements in sequence rather than treating them as one claim.

The resulting answer is bounded and consistent. Middle-layer representations carry a measurable language-training signal under the recorded controls, but plain distillation's alignment loss remains entangled with language quality; the tested alignment objectives do not establish a reliable participant-level gain; capacity, loss-form, voxelwise, and full-fine-tuning variants do not rescue the intervention; and neither the tested downstream routes nor the dense synthetic proxy demonstrates real-brain benefit [E002] [E003] [E006] [E008] [E009] [E011] [E013] [E016] [E017] [E024]. These findings do not imply that neural supervision is universally ineffective. They show that brain alignment is a valid diagnostic under explicit controls but is not a demonstrated reliable training signal in the compression regimes studied here.

The durable contribution is a validation standard for future brain-guided training. Language quality, target specificity, inference unit, and downstream transfer must be separated because each can produce an apparently positive result without establishing the next link in the argument. A model that fits a neural target better has not necessarily become more aligned with measured brains; a model that scores higher on an averaged response has not necessarily improved across individuals; and a model with a changed representation has not necessarily become more useful. Treating measurement, intervention, biological generalization, and payoff as distinct estimands turns brain alignment from an attractive scalar score into a testable scientific claim.

# A Experiment coverage and disposition

This appendix accounts for every experiment identifier used by the project, including engineering pilots, unrun designs, failed precondition gates, and intentionally unused identifiers. An experiment is classified as scientific evidence only when it produced an interpretable result for a stated estimand; implementation activity alone is not treated as evidence. Table 6 gives the complete disposition, while Appendix B expands the secondary experiments and failed instruments that would interrupt the main causal sequence.

**Table 6: Complete experiment coverage and disposition.**

| ID | Question | Execution state and verdict | Location and boundary |
|---|---|---|---|
| E001 | Recover a planted synthetic relationship? | Implemented; engineering validation only. | Appendix B.1; no real-neural claim. |
| E002 | Predict Tuckute responses beyond nuisance and untrained features? | Complete; controlled ROI measurability pass. | Section 4.1; raw unique $R^2$, no invalid ceiling fraction. |
| E003 | Retain alignment under distillation? | Complete; quality-entangled retention gradient. | Section 4.2; objective-specific shedding unidentified. |
| E004 | Move held-out alignment with a brain loss? | Complete and reanalysed; no demonstrated lever at five-fold inference. | Section 4.3; 15-cell bootstrap superseded. |
| E005 | Beat a matched-perplexity control on an averaged target? | Complete and reanalysed; small averaged-target trend. | Section 4.3; Appendix C.4; original excludes-zero claim superseded. |
| E006 | Extend measurability to naturalistic voxels? | Complete and corrected; conditional UTS03 pass. | Section 4.1; voxel-population intervals and derived MDE retracted. |
| E007 | Power calculation following E006? | Identifier intentionally unused. | No experiment record; invalid calculation never promoted. |
| E008 | Generalize the intervention across participants? | Complete; participant-level null for the predeclared effect. | Section 4.3; Appendix C.3; nine participant units. |
| E009 | Improve out-of-domain perplexity? | Complete; bounded practical null with an unreliable mediator. | Section 4.5. |
| E010 | Increase the contrast by averaging more participants? | Complete including random-subset rerun; no identified dose response. | Section 4.3; Appendix B.2; nested interpretation superseded. |
| E011 | Rescue the result with greater adapter capacity? | Complete; higher rank and duration did not strengthen the manipulation. | Section 4.4. |
| E012 | Run a voxelwise participant-level study? | Deferred and unrun because three participants could not support the claim. | Coverage only. |
| E013 | Rescue the intervention with contrastive or voxelwise objectives? | Complete for tested branches; ROI null and voxelwise manipulation failure. | Section 4.4; larger expansion unbuilt. |
| E014 | Improve encoding by averaging without intervention? | Complete; legitimate SNR improvement and estimand shift. | Section 4.3; Appendix B.3; no universal artifact claim. |
| E015 | Relate model quality to alignment? | Complete and corrected; strong full-range, moderate uncertain capable-band coupling. | Section 4.2; Appendix C.5; $r \simeq -0.92$ superseded. |
| E016 | Train on a dense synthetic neural target and transfer? | Complete and independently reviewed; synthetic positive, fixed real-brain transfer negative. | Section 4.6; Appendix B.4; dimension-matched control and seed-unit scope. |
| E017 | Reveal a lever with full fine-tuning? | Complete; three-participant existence probe found no specific gain. | Section 4.4; Appendix C.6; no nine-cell population inference. |
| E018 | Reserved identifier? | Intentionally unused. | No scientific content. |
| E019 | Reproduce a local Negi-style gain? | Complete; corroborative local lever failure. | Section 5.6; Appendix B.5; not a Negi refutation. |

| ID | Question | Execution state and verdict | Location and boundary |
|---|---|---|---|
| E020 | Estimate a stimulus-predictable residual ceiling? | Complete; failed instrument with bounded controlled residual. | Section 4.6; Appendix B.6; full ceiling remains open. |
| E021 | Is reading-time benefit cognition-specific? | Complete through phase-randomized control; shape-specific, not cognition-specific. | Section 5.6; Appendix B.7; earlier reads superseded. |
| E022 | Reproduce a Moussa-style speech gain? | Complete pilot; reduced single-participant non-reproduction. | Section 5.6; Appendix B.8; not a full-study refutation. |
| E023 | Identify objective-specific shedding against a quality frontier? | Design plus analysis-only slope gate; no training result. | Appendix B.10; decisive saturated comparison not cleared. |
| E024 | Use gaze as train-only privileged information? | Gaze preconditions complete; intervention stopped for informativeness and split validity. | Section 4.5; Appendix B.9; EEG untested. |

### A.1 Supersession map

Five corrections materially change how earlier results are described. First, E004 and E005 originally treated seed–fold cells as independent observations; current inference is performed at the fold unit, and E008 supplies the participant-level test. Second, E006 originally bootstrapped thousands of spatially dependent voxels as if they were independent biological replicates; the retained result is a conditional one-participant point contrast with five held-out-block sensitivity checks. Third, E010's nested participant averages suggested a monotone dose response, but the E010b random-subset analysis showed that subset identity and target construction could explain the pattern. Fourth, E015's initial eight-model correlation of approximately $-0.92$ was replaced by a tokenizer-independent 22-model, six-family analysis with a weaker capable-band relationship and an effective-generation limitation. Fifth, E021's apparent reading-time advantage survived ordinary learnability controls but disappeared against a content-free control matched for autocorrelation and marginal shape; the final claim concerns signal shape rather than cognition.

The Tuckute noise-ceiling correction is a scale correction rather than a changed experimental result. The stored value 0.491 is a correlation ceiling, whereas unique $R^2$ is on a variance scale, so earlier ratios of unique $R^2$ to 0.491 are invalid and are not used here. [E002] The raw unique-$R^2$ values, trained-minus-untrained contrasts, intervention contrasts, and their substantive verdicts are unchanged.

## B Secondary experiments and failed instruments

This appendix explains experiments that validate the apparatus, diagnose a main-track result, test a failed measurement instrument, or provide a reduced external reproduction. Each subsection states the question, intervention or comparison, estimand and inference unit, result, conclusion, limitation, and role in the thesis.

### B.1 E001: end-to-end implementation validation on a planted synthetic signal

E001 asked whether the initial implementation could recover a known relationship from synthetic language features and synthetic fMRI-like targets. The run exercised feature extraction, ridge fitting, cross-validation, and result serialization while holding the planted data-generating relationship under experimenter control. Its estimand was recovery of that constructed relationship, and the relevant unit was the synthetic sample rather than a participant or brain measurement. The implementation recovered the planted signal and therefore passed its engineering validation. [E001] This result licenses use of the implementation as a starting point, but it provides no evidence that a language model predicts real neural activity, that the nuisance controls are sufficient on real stimuli, or that a neural objective can improve training. E002 is the first scientific evidence because it replaces the constructed target with recorded language-network responses and adds trained-versus-untrained and nuisance controls.

### B.2 E010 and E010b: participant averaging as a target intervention

E010 asked whether the apparent brain-specific intervention contrast increased as more participants were averaged into the target. The motivation was the discrepancy between E005, which used a five-participant average, and E008, which found a near-zero mean across individual participants. For nested averages of $k \in \{1, 2, 3, 5, 9\}$ participants, the measured intervention gaps were approximately $-0.0002, -0.0001, +0.0023, +0.0194, +0.0071$. [E010] The original interpretation treated the rising $k = 1$-to-5 limb as a reliability dose response while noting the reversal at $k = 9$. That interpretation was not identified because every larger target contained the smaller target, participant order was fixed, and target construction changed together with signal-to-noise ratio.

E010b therefore repeated the analysis over random participant subsets at each $k$. The resulting subset distributions overlapped substantially and did not support a stable monotone causal relation between participant count and the intervention contrast. [E010] The estimand in both analyses was the contrast for a constructed group-average endpoint, while participant subsets, not seed–fold cells, supplied the sensitivity variation. The joint conclusion is that averaging changes both reliability and the biological target, and the nested curve does not establish that increasing $k$ causes a brain-specific training effect. Its role is diagnostic: it explains why E005 and E008 can differ without treating either endpoint as universally correct.

### B.3 E014: averaging improves encoding signal without establishing an intervention artifact

E014 separated the effect of participant averaging on the encoding measurement from its effect on brain-guided training. It held the model features and encoding procedure fixed while replacing individual targets with averages over increasing numbers of participants. Raw encoding unique $R^2$ rose across $k = 1, 2, 3, 5, 9$ as $+0.0039, +0.0162, +0.0324, +0.0444, +0.0702$. [E014] This is the expected consequence of suppressing participant-specific and measurement noise in a shared stimulus-evoked target. The estimand is nevertheless different at each $k$: an individual response at $k = 1$ and an increasingly group-shared response as $k$ grows. E014 therefore supports participant averaging as a legitimate way to improve the signal-to-noise ratio of a group-level encoding target. It does not show that averaging universally manufactures a false encoding score, and it does not rescue participant-level inference for E005. Its role is to keep the SNR benefit separate from the biological-generalization question answered by E008.

### B.4 E016 prerequisite gates and the synthetic-to-real transfer test

E016 asked whether a dense stimulus-predictable neural proxy could overcome the scarcity of measured fMRI targets and provide a usable auxiliary training signal. The program used TRIBE v2 to generate a high-dimensional synthetic neural target from text-associated stimuli, trained language-model students with that target, and then evaluated whether the resulting improvement transferred to an independent real-brain endpoint.

The first prerequisite was fidelity: a synthetic target is useful only if it captures stable stimulus-dependent structure and can be predicted by the selected student representation. The fidelity analyses showed that the proxy was learnable enough to support a controlled training experiment, but a separate attempt to use it as a formal ceiling on real-brain alignment failed because proxy predictions captured only a limited and target-dependent portion of recorded response variance. [E016] The ceiling analysis is therefore a failed instrument rather than evidence that no additional stimulus-predictable neural structure exists.

The training comparison used six stochastic seeds and the same KD backbone across target arms. On their respective synthetic endpoints, the TRIBE-guided arm improved target $R^2$ over KD by +0.078298, whereas the sentence-local projected teacher-feature control improved by +0.000903. [E016] The text-feature target matched output dimension but not information content, geometry, baseline difficulty, or effective rank, so the comparison licenses only the statement that TRIBE outperformed this particular dimension-matched control on its own synthetic endpoint. The block-permuted control also retained some fixed points and is described as block-permuted rather than fully deranged.

The independent transfer gate scored saved students on the fixed Tuckute target averaged over five participants and five functional ROIs. The inference unit was the six stochastic training seeds for this one endpoint, not participants. TRIBE-minus-text-feature gain relative to KD was $-0.001213$; all six margins were negative and the exact sign test gave $p = 0.03125$. [E016] The corresponding comparison against permuted-control gains was also negative, and PCA ranks 25, 50, and 100 did not reverse the route. Synthetic training used layer 6 while the Tuckute diagnostic read layer 7, which further bounds mechanistic interpretation. The conclusion is a transfer failure in this fixed diagnostic, not a population-level null and not a universal upper bound on synthetic neural supervision. E016's role is decisive for the thesis logic: learnability of a neural-looking target does not establish usefulness for an independently measured biological endpoint.

## B.5 E019: external Negi-style objective under local controls

E019 asked whether a faithful-as-feasible implementation of a published Negi-style brain-tuning head would first reproduce a local gain and then survive matched-quality and target-specificity controls. The implementation combined the available naturalistic LeBel substrate with a differentiable temporal head and tested learning-rate, probe, and positive-control variants before the final comparison. The relevant estimand was held-out alignment change for the real target relative to a permuted target under the same training and evaluation path. The head did not produce a stable target-specific lever on the local substrate, so the planned matched-quality collapse of a reproduced positive could not be performed. [E019] Across three participants and three training seeds, gentle tuning changed encoding correlation by $-0.0006 \pm 0.0027$ and unique $R^2$ by $-0.0001 \pm 0.0010$ relative to the base model, where the terms after $\pm$ are descriptive standard deviations across the nine participant–seed cells. [E019] The real-minus-permuted specificity contrast was $-0.0004 \pm 0.0018$, and the real-minus-intended-quality-control contrast was $-0.0010 \pm 0.0018$ on the same descriptive cell scale. [E019] The intended quality control did not achieve a match: the real-target arm had perplexity about 51.1–51.7, while the control was about 40.1–40.3, so the latter contrast is not a matched-quality estimate. [E019] Participants, not the nine participant–seed cells, are the biological inference units; these values are mechanism checks rather than population confidence intervals. Stronger learning rates produced negative encoding changes, and the strongest probe increased perplexity from approximately 50 to 11,514 while reducing encoding correlation by about 0.055. [E019] This corroborates the mechanism failure seen in the thesis's own objectives because a distinct objective family also failed to establish movement. It does not reproduce the published dataset, participant pool, full training scale, or exact implementation, and therefore cannot refute Negi et al. Its role is convergent boundary evidence: the local null is not confined to one MSE loss, but external validity remains unresolved.

## B.6 E020: an empirical stimulus-predictable ceiling that did not close

E020 asked whether an empirical reference constructed from other participants could approximate $\mathbb{E}[Y \mid S]$ and reveal brain response structure beyond the stimulus-predictable mean. The intended comparison residualized a target participant's response against a leave-one-participant-out group reference, then measured how much language-model signal remained. An initial residual appeared to preserve substantial alignment, which would have suggested a participant-specific component beyond the shared stimulus response. Diagnostics showed that the reference was itself unreliable (0.33), residual split-half reliability was 0.174, and temporal leakage and autocorrelation could preserve apparent signal after subtraction. [E020] Once the cleanest controlled construction was used, the trained-minus-untrained residual gap was $-0.018$. [E020] The inference used six participants on one fixed story, and the reference did not provide a high-reliability estimate of the latent conditional mean. E020 therefore rejects this particular ceiling instrument while leaving the general ceiling question open. Its role is to prevent the Discussion from turning a failed measurement into either a positive participant-specific mechanism or a universal closure result.

## B.7 E021: reading-time content versus statistical shape

E021 asked whether human reading time supplies a cognition-specific auxiliary target after removing the component predictable from language-model surprisal. The early comparisons used raw reading time, surprisal-residualized reading time, permuted targets, random targets, word length, frequency, and other-model surprisal, and evaluated frozen-probe learning curves across training seeds. Raw and surprisal-residualized reading time performed similarly, showing that removal of the model's own surprisal-predictable component did not remove the apparent benefit. [E021] In the penultimate analysis the residual target beat several learnability-matched controls, but those controls did not match its strong temporal autocorrelation and heavy-tailed marginal distribution. The final control phase-randomized the signal while preserving its autocorrelation and approximate marginal shape but removed behavioral content. That content-free shape twin tied the reading-time residual: the difference in area under the frozen-probe learning curve was $+5.3$ with a 95% interval $[-7.9, +18.5]$, where lower area means faster or better learning. [E021] The independent units were stochastic training seeds, and the claim is restricted to this learning-curve setup and target construction. The conclusion is that the observed advantage is attributable to signal shape or regularization, not demonstrated cognition-specific information. E021's role is methodological: auxiliary targets require controls matched for learnability and statistical geometry, not only random or permuted labels.

## B.8 E022: reduced Moussa-style speech reproduction

E022 asked whether brain-guided tuning of a pretrained speech representation reproduced a reported gain in phoneme classification before investing in a full matched-quality shrink test. The reduced setup used one LeBel participant, 24 stories, a wav2vec2-base initialization for both arms, and three training seeds. The primary estimand was the brain-tuned minus pretrained phoneme-F1 difference on held-out speech. The observed gain was $+0.52$ F1 points with a 95% interval $[-0.36, +1.40]$, below the predeclared $+2$-point continuation gate and including zero. [E022] The planned matched-twin shrink test was consequently vacuous and was not built. Because this setup is smaller and differs from the published multisubject experiment, it is a reduced non-reproduction rather than evidence that the published effect is false. Its role is to show that the thesis did not obtain an external positive on which to demonstrate a matched-quality collapse.

### B.9 E024: privileged gaze information stopped at the precondition gate

E024 asked whether gaze or EEG observed only during training could improve the sample efficiency of a text classifier, following the learning-using-privileged-information setting. The planned estimand was the area between held-out learning curves for a text-only student and privileged-information arms at paired training subsets. Before building five training arms, the design predeclared gaze as the first binding gate and required it to be reliable, informative about the label under natural reading, and evaluable with a leakage-free grouped split.

The reliable gaze summaries largely tracked sentence length and did not predict the relation label under natural reading. Across binary and multiclass probes, linear and nonlinear readouts, participant-averaged and within-participant features, natural-reading performance stayed near chance; the rich multiclass probe gave balanced accuracy 0.202 against chance 0.200 with permutation $p = 0.35$. [E024] In task-directed reading, by contrast, gaze predicted relation type with balanced accuracy 0.295 against chance 0.100 and permutation $p = 0.003$. [E024] This contrast indicates task-intent leakage rather than a portable natural-reading signal.

The dataset also contained 300 sentences drawn from only seven paragraphs, so a paragraph-disjoint test set varied from 8 to 161 sentences across split seeds and did not support a stable low-data learning-curve estimand. These two failures were independent: the signal was uninformative for the clean target, and the grouping structure prevented a stable leakage-free evaluation. The five-arm LUPI intervention was therefore stopped at its binding gaze precondition gate, and EEG was not evaluated. This is not a completed negative intervention or an EEG precondition result; it is evidence that the proposed gaze-first substrate cannot identify the intended transferable privileged-information effect.

### B.10 E023: objective-specific shedding as an unexecuted identification program

E023 formalized a counterfactual question left open by E003 and E015: whether a distilled model lies below the alignment expected for a normally trained model of the same language-model quality. It proposed a same-lineage quality frontier, a matched-quality KD-versus-ordinary-fine-tuning comparison, and operation in a regime where the frontier is sufficiently flat to separate objective effects from ordinary quality loss. The only executed component was an analysis-only slope and knee gate using already measured Pythia and Qwen models. The Pythia 1B-to-410M pair met the local flatness screen with slope $+0.015$, alignment change $+0.0007$, and quality change 0.066 bits per byte. [E023] The three checked Qwen pair slopes were $+0.334$, $-0.217$, and $+0.076$, so none cleared the decisive saturated-regime requirement. [E023] The analysis-only gate marked a lower-quality pilot as feasible but did not clear a decisive training comparison; no inferential unit or hypothesis test was claimed for this descriptive screen. No KD-versus-fine-tuning intervention was run, so E023 contributes an identification framework rather than a scientific result about objective-specific shedding.

## C Formal and statistical detail

### C.1 Notation and the assumption-bound relationship between conditional information and unique variance

Let $Y$ denote a scalar brain-response target, $X$ the language-model feature block, and $Z$ the nuisance feature block. Conditional mutual information asks how much statistical dependence remains between $X$ and $Y$ after $Z$ is known:

$$I(X;Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z). \tag{12}$$

This definition is distributional and does not specify a linear estimator. Under a population multivariate Gaussian model with correctly specified unregularized conditional means, the reduction in conditional residual variance can be written as

$$I(X;Y \mid Z) = -\frac{1}{2}\log\left(1 - R^2_{\text{partial}}\right). \tag{13}$$

Here $R^2_{\text{partial}}$ measures the fraction of the variance left after $Z$ that is explained by adding $X$.

The empirical score used in this thesis is instead semipartial cross-validated unique variance,

$$\Delta R^2 = R^2_{\text{full}} - R^2_{\text{nuisance}}. \tag{14}$$

Under the same population assumptions, partial and semipartial coefficients are related by

$$\Delta R^2 = (1 - R^2_{\text{nuisance}})R^2_{\text{partial}}. \tag{15}$$

The thesis fits regularized ridge models on finite folds and evaluates them out of sample, so neither equality makes the measured $\Delta R^2$ an estimator or lower bound for conditional mutual information. The information-theoretic formulation supplies a conceptual target, while the actual estimand is incremental held-out linear prediction beyond the specified nuisance design.

The nuisance set is also necessarily incomplete. Length, position, static embeddings, acoustic tiers, and temporal controls block named alternative pathways, but they do not condition on every aspect of stimulus content. A positive $\Delta R^2$ therefore licenses the statement that contextual features add held-out linear predictive value beyond those controls. It does not isolate a neural computation, establish causal equivalence, or distinguish prediction of a shared high-level stimulus component from prediction of participant-specific processing.

## C.2 Regularization, data processing, generalization, and the rate–distortion analogy

An auxiliary brain term gives an optimization objective of the form

$$\widehat{\theta} = \arg\min_{\theta} \left\{\mathcal{L}_{\text{text}}(\theta) + \lambda\mathcal{L}_{\text{brain}}(\theta)\right\}. \tag{16}$$

The brain term biases optimization among solutions that also fit the text objective. It can be read as a maximum-a-posteriori prior only if $\exp[-\lambda\mathcal{L}_{\text{brain}}(\theta)]$ is normalizable and independent of the observed text likelihood. Without those conditions it is more precise to call it an auxiliary loss or regularizer.

The data-processing inequality states that $I(A;C) \leq I(A;B)$ for a justified Markov chain $A \to B \to C$. If a compressed representation $C$ is literally a fixed post-processing of teacher representation $B$, then its information about a brain variable $A$ cannot exceed the teacher's. Ordinary knowledge distillation does not automatically instantiate this chain because both teacher and student receive direct stimulus or corpus information. The inequality therefore motivates a preservation question but does not predict the sign of the empirical encoding-score change under training.

For a learning algorithm mapping a sample $Z^n$ to parameters $W$, a sub-Gaussian information-theoretic generalization bound has the form

$$\left|\mathbb{E}[L(W) - \widehat{L}(W)]\right| \leq \sqrt{\frac{2\sigma^2 I(W;Z^n)}{n}}. \tag{17}$$

The right-hand side grows with algorithm–sample mutual information. The bound does not imply that adding a neural signal creates a benefit, that the benefit scales as an added information budget, or that a low-data regime must improve. An auxiliary constraint could tighten this bound only if it reduces dependence on the sampled text while preserving empirical fit, which was not measured here. Earlier order-of-magnitude information-budget calculations are therefore not part of the thesis argument.

Classical rate–distortion theory defines

$$R(D) = \min_{P_{\widehat{S}|S}:\mathbb{E}[d(S,\widehat{S})]\leq D} I(S;\widehat{S}). \tag{18}$$

The empirical experiments do not identify this object because parameter count or FLOPs are not the mutual-information rate and alignment loss is not established as the source-coding distortion. The phrase "rate–distortion-style trade-off" is used only for the engineering Pareto relation among model budget, language-model quality, and measured alignment.

## C.3 Inference units and power

An inference unit is the independently varying entity over which an uncertainty statement generalizes. Training seeds quantify optimizer variability, contiguous test folds quantify sensitivity to stimulus blocks, voxels quantify spatial measurement breadth, and participants quantify biological population variation. These units answer different questions and cannot be substituted by multiplying their counts.

E004 and E005 used three seeds and five rotating folds, producing 15 observed seed–fold cells. Because seeds within one fold share the same held-out stimulus block and fold effects dominate the contrast, treating all 15 cells as independent understates uncertainty. E004's mean Qwen contrast was +0.00316, but the five-fold $t$ interval was [−0.00227, +0.00858], and leaving out the influential fourth fold reduced the mean to +0.0014. [E004] The correct conclusion is no demonstrated fold-level lever, with limited power to exclude a small effect.

E008 instead averaged repeated training variability within each of nine participants and treated participants as the biological inference unit. Its mean was +0.00010 with participant-axis $t$ interval [−0.00037, +0.00058]. [E008] The predeclared effect of +0.003 lay well outside that interval, providing high power against the effect size the study was designed to detect while leaving smaller effects possible. [E008]

E006 illustrates a distinct error. Bootstrapping 11,442 spatially dependent voxels generated a narrow interval but did not create participant replication. The retained evidence is the conditional UTS03 point gap, +0.0207 for GPT-2 and +0.0277 for Qwen, plus positive signs on all five held-out story blocks. [E006] Because those blocks share training stories, even the five signs are sensitivity evidence rather than an independent population confidence interval.

Power calculations must use the variance of the future paired contrast at its intended inference unit. The retracted E007 calculation used unmatched absolute fold summaries from E006 and therefore did not estimate the variance of a future tuned-minus-control effect. No replacement participant count is inferred from it.

## C.4 E005 reanalysis and the lambda sweep

E005 compared real-target brain-guided KD with a block-permuted-target twin while matching the text-training setup and approximately matching perplexity. The initial averaged-target estimate was +0.0081. [E005] Its original bootstrap sampled 15 seed–fold cells independently and excluded zero, but the corrected fold-unit analysis showed that one held-out block carried a disproportionate share of the result and the valid interval included zero. The retained description is therefore a small averaged-target trend rather than a confirmed brain-specific improvement.

A later Qwen lambda sweep asked whether stronger auxiliary weighting produced a stable alignment gain without unacceptable language-model degradation. Larger weights moved the training endpoint more strongly but also worsened perplexity, and no setting established a free target-specific improvement at matched quality. [E005] This supports a trade-off description rather than a favorable frontier shift. The sweep is contextual because it reuses the participant-averaged target and does not replace E008's participant-level test.

### C.5 E015 family, band, and partial-correlation sensitivities

E015 scored 22 decoder language models from six named families using tokenizer-independent bits per byte and the same controlled Tuckute alignment endpoint. Across the full capability range, the Pearson relationship was approximately $r = -0.78$. [E015] Removing near-floor models weakened the association: in the capable band it was approximately $-0.50$, and in the overlapping quality band approximately $-0.48$. [E015] The Fisher intervals for these band-restricted correlations included zero because the bands contained only eight to ten models.

A partial correlation controlling for log parameter count was $-0.675$ with Fisher 95% interval $[-0.857, -0.344]$. [E015] This is not a clean decomposition into scale and quality because bits per byte and scale were strongly collinear and the residual variation also encoded family, data, and architecture differences. Family-cluster resampling remained negative, but the six families separated into roughly two model generations, limiting the effective independent breadth. Participant-level analyses showed the pattern in the one participant with adequate alignment signal and were uninformative in two participants near the measurement floor. The durable conclusion is that quality is a serious confound requiring matched-quality controls, not that perplexity deterministically explains brain alignment within the capable regime.

### C.6 E017 intervention sequence and E023 identification gate

E017 tested whether full fine-tuning could reveal movement that low-rank adaptation missed. An aggressive configuration caused catastrophic forgetting and failed the language-model-quality requirement. A gentler UTS03 pilot showed that the representation could move without the same failure, which cleared a three-participant existence test. Across UTS01, UTS02, and UTS03, the real-target arm did not improve over its target-specific controls. [E017] The observed subject–seed cells share only three biological units, so the result is reported descriptively as a three-participant mechanism probe rather than with a population confidence claim.

E023 then asked how an objective-specific loss could be separated from the alignment expected at the student's quality. Its proposed estimand was the vertical residual of a KD student below a same-lineage alignment–quality frontier, paired against ordinary fine-tuning from the same initialization. The analysis-only slope gate found one locally flat lower-quality Pythia pair but no decisive saturated Qwen regime. [E023] Because the identifying region was not established, the training comparison was not launched. Together E017 and E023 distinguish a tested full-fine-tuning null from an untested larger-scale counterfactual program.

### C.7 Compact notation

| | |
|---|---|
| $S$ | Language stimulus, such as a sentence or story segment. |
| $X$ | Language-model feature block for the stimulus. |
| $Y$ | Recorded or synthetic target response. |
| $Z$ | Specified nuisance feature block. |
| $\Delta R^2$ | Cross-validated semipartial unique $R^2$, the full-model $R^2$ minus nuisance-model $R^2$. |
| $\mathcal{L}_{\text{brain}}$ | Auxiliary loss relating a student representation to a neural or cognitive target. |
| $\mathbb{E}[Y \mid S]$ | Stimulus-predictable conditional mean of a response. |

**ROI** Region of interest, a prespecified group or aggregate of voxels.

**BPB** Bits per byte, a tokenizer-independent language-model quality measure on a fixed corpus.

**KD** Knowledge distillation, training a student to match a teacher's softened output distribution.

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